

ROLE OF URBAN GREEN SPACES IN RESIDENTIAL NEIGHBOURHOODS, HYDERABAD

PLANNING THESIS

ON

**‘ROLE OF URBAN GREEN SPACES IN RESIDENTIAL
NEIGHBOURHOODS, HYDERABAD’**

(A Case of Jaya Prakash Narayan Road Stretch)

Prepared In Partial Fulfilment of the Requirements for the Award Of

**BACHELOR OF TECHNOLOGY
(URBAN AND REGIONAL PLANNING)**

By

B. NIKHIL

17011BA004

Under The Supervision Of

MS. P. DHRAUVITHA



JAWAHARLAL NEHRU ARCHITECTURE & FINE ARTS UNIVERSITY

**SCHOOL OF PLANNING & ARCHITECTURE
DEPARTMENT OF URBAN & REGIONAL PLANNING
HYDERABAD, TELANGANA - 500 028**

2021



SCHOOL OF PLANNING AND ARCHITECTURE, JNAFAU

Mahavir marg, Hyderabad – 500 028

DEPARTMENT OF URBAN & REGIONAL PLANNING

CERTIFICATE

I/We certify that the Final Project entitled

**ROLE OF URBAN GREEN SPACES IN RESIDENTIAL
NEIGHBOURHOODS, HYDERABAD'**

Submitted by Mr. **B. NIKHIL** bearing Roll No. **17011BA004** on **28/07/2021**
in partial fulfillment of the requirements for the award of the degree of
Bachelor of Technology (Planning) of this University is a bonafide work to
the best of my/our knowledge and may be placed before the Examination
Board for their consideration.

Ms. P. Dhruvitha
Project Guide

Mr. Siddhartha
Project Coordinator

Mrs. E. Sandhya Reddy
Head of the Department, URP

External Examiner

Dr. S. Kumar
Principal, SPA

DECLARATION

I hereby declare that the thesis entitled “**ROLE OF URBAN GREEN SPACES IN RESIDENTIAL NEIGHBOURHOODS, HYDERABAD**” which is submitted for the partial fulfillment of degree of bachelors in Urban and Regional planning of the Jawaharlal Nehru Architecture and Fine Arts University, Hyderabad, is the result of work carried out by me in the School of Planning and Architecture. I worked under the guidance of ***Ms. P. Dhruvitha***

ACKNOWLEDGEMENT

Many people deserve recognition for their support and guidance during my studies. I am especially indebted to my Guide, Ms. Druvitha, Assistant professor for Urban and Regional Planning, who has been guiding me for my thesis study and research. Her patience and understanding, patience was extraordinary. Without her guidance my vision for this study would not have become a reality.

I would like to express my deepest appreciation to all those who provided me the possibility to complete the thesis.

Furthermore, I would also like to acknowledge with much appreciation the crucial role of all the faculty members, Department of Urban and Regional Planning for their constructive inputs and support.

I am highly indebted to Mrs. E. Sandhya Reddy, Head of the Department, and Mr. Siddhartha, Thesis Coordinator for their guidance and constant supervision.

I would like to acknowledge my parents for supporting, encouraging me throughout my thesis. They deserve special recognition for their support and prayers during this long journey to complete my studies. Thanks to my parents, and brothers and sister for the support. Their support and inspiration gave me the stamina to be more diligent in my work, and follow my research schedule to complete my study on time. My special thanks to my friends for the keen interest shown in helping me to complete this thesis successfully.

ABSTRACT

Just like water, drainage, electricity, education and healthcare are considered essential public services, the role of Green spaces is equally important in establishing and maintaining quality of life in a community. They contribute to health and environmental wellbeing of a community and a region. There are different types of activities associated with different age groups in a park. The physical elements and activities in the urban green spaces offer many benefits to the citizens, namely health, social interaction and economic value. They provide identity and sense of place to the citizens.

This paper explains the benefits and challenges of urban green spaces based on the critical discussion of study results from different studies in different cities. The important roles played by green spaces are social, economic, cultural and environmental aspects of sustainable development. Urban green spaces can be a comprehensive tool for long term protection of environmental sustainability through improving the quality of life and air quality, increasing property value due to their amenity and aesthetic characteristics, and reducing the energy costs of cooling buildings. Urban green spaces also can provide ecosystem services in which the recreation and relaxation facilities are especially available to urban dwellers and tourists too. To confirm the multiple roles played by green spaces, certain level of qualitative improvements and distribution of green spaces within the urban area should be considered and incorporated effectively into the environmental sustainability agenda. To do this, an integrated approach regarding the planning, monitoring, designing and maintaining of urban green spaces is required for improving the environmental sustainability in cities

Table of Contents

1. INTRODUCTION	2
1.1 BACKGROUND STUDY	4
1.2. The Urban Environment	6
1.3. Need of urban green space in present world.....	7
1.4. Concept of Urban Green Spaces:	7
1.5. Urban Planning.....	9
1.6. Green spaces in cities	10
1.7. CHARACTERISTICS OF URBAN GREEN SPACES:	13
1.8. IMPORTANCE OF GREEN ENVIRONMENTS FOR RESIDENTIAL COMMUNITIES:	14
1.9. GREEN SPACE FUNCTIONS AND BENEFITS:	15
1.9.1. General benefits:	15
1.9.2. Benefits of Green spaces: The Ecosystem Services Perspective	16
1.9.3. What do Good Urban Green Spaces Offer?.....	21
1.10. Social Benefits	35
1.11. Challenges Facing in urban green spaces	38
1.12. Basic Requirements for Developing an Urban Greening Program.....	41
1.13. URBAN GREEN GROWTH STRATEGIES FOR INDIAN CITIES	42
1.14. MAJOR INDIAN CITIES WITH PER CAPITA GREEN SPACE.	43
2. DESKTOP STUD	45
2.1. International desktop study	45
2.2. History of Green Spaces in Madrid:.....	45
2.2.1. Green Spaces in planning practices in Madrid:.....	46
2.3. Green Spaces in Stockholm:.....	48
2.4. History of Green spaces in Stockholm	49
2.4.1. Results obtained from Madrid	53
2.4.2. Results obtained from Stockholm	54
2.4.3. Analysis of results of Madrid.....	54
2.4.4. Analysis of results of Stockholm	55
2.2. A CASE STUDY OF PUNE, INDIA:.....	57
2.2.1. Landuse/Landcover Changes in Pune City	58
3. STUDY AREA	60
3.1. PROFILE:.....	60
3.2. DEMOGRAPHIC PROFILE OF STUDY AREA:	60

3.3. Timeline of the study area SAIDABAD 2000 – 2021 (10 years)	62
3.4. STUDY AREA JUSTIFICATION:.....	65
3.5. ANALYSING CLASSIFICATIONS OF PUBLIC URBAN GREEN SPACE ...	66
3.6. GREEN SPACE INDEX FOR GHMC	67
3.7. SWOT ANALYSIS	70
4. RECOMMENDATIONS	71
Green Roofs	71
Green streets.....	71
Land conservation	72
BIBLIOGRAPHY	73

List of Tables

Table 1: Buffer distance	53
Table 2: Total area and % of Green spaces	53
Table 3: Total area and % of green spaces Stockholm	54
Table 4: Land use Land cover changes in pune city	57
Table 5: Demographic Data of study area	61
Table 6: PERCENTAGE OF PARKS WITHIN ZONES OF GHMC	65
Table 7: Percentage of green	66
Table 8: Land use percentage for different developed areas	68
Table 9 - Study area land use percentage	68
Table 10 Comparison of green space with URDPFI	69
Table 11 Planning Norms, Standards for recreational spaces	69

List of Figures

Figure 1: CHARACTERISTICS OF URBAN GREEN SPACES.....	13
Figure 2: GREEN SPACE TYPES AND FUNCTIONS	15
Figure 3: Healthier Community	20
Figure 4: Benefits of Green spaces	20
Figure 5: Benefits of Urban Green spaces	23
Figure 6: Urbanization in India	43
Figure 7: Per capita green spaces	44
Figure 8: Madrid city Map	45
Figure 9: Urban green areas in Madrid	46
Figure 10: Stockholm District Map.....	48
Figure 11: Urban Green spaces in stockholom	49
Figure 12: Buffer areas around urban spaces.....	53
Figure 13: Buffer areas around urban spaces in stockholm	54
Figure 14: Location map of study area	60
Figure 15: 2000	62
Figure 16: 2010	62
Figure 17: 2020	63
Figure 18: Present state of the study area SAIDABAD ward of year 2021.....	63

1. INTRODUCTION

Since the beginning of Mankind, nature has provided humans with shelter, resources, etc., being an important part of their development. Although this relationship has changed and evolved from one of the main aspects in human life to areas dedicated to recreation and similar activities, the relationship between humans and nature is still present. This relationship is easily manifest in rural areas, where the population has a direct relationship with the nature, either by direct access to it, to the growth of food, etc.

But in urban areas, this relationship is more limited, and it usually is directly related to the presence of green spaces in the city. Moreover, these green spaces have traditionally been considered as 'good places' for people. By this their presence and importance in cities are understandable. There are multiple definitions of green spaces. They define green space as an "open, undeveloped land with natural vegetation", a definition that includes areas such as parks, forests, vegetation corridors, playing fields, wetlands, urban forests, cemeteries etc.

This thesis in regards of green spaces, the main focus has been put into to investigate if there are any benefits of green spaces, as well as in articles reflecting upon /urbanization processes and green spaces. The studies reflecting upon the relationship between green spaces and the benefits they have on human population has been increasing lately, as an answer to concern about issues produced by urbanization, in relation to the environment, and changes in human lifestyles. The relationship between the presence of green spaces and their positive effects was done through articles that have researched. Literature about green spaces and their benefits was wide, but was found somehow incomplete, as most of the articles related to a limited number of effects of these green spaces, but most of them fail to do a complete study.

Urbanization results in an increasing proportion of the population living in cities. Urban living limits access to nature and can increase exposure to certain environmental hazards, such as air and noise pollution. Many urban areas face increasing pressure from expanding populations, limited resources and growing impacts of climate change. These challenges must be addressed in order for cities to provide healthy and sustainable living environments.

Although it has great many dimensions, ranging from physical to socio- cultural, psychological and environmental ones, the role of economic factors is undeniably decisive in raising its level. In a presentation which will be dealing with the linkages between environmental conditions and the quality of life, the concept of environmental needs to be regarded in its broadest sense, covering not only physical, but also socio-cultural, political and economic requirements of human prosperity. In order to analyse the relationships between the quality of life and environmental conditions adequately, it would be appropriate to have a brief look at urban and environmental stresses affecting the rapidly urbanizing world at the beginning. Rapid population growth and urbanization, industrialization and technological change, increased expectations caused by rising affluence, gaps between the level of expectations and actual consumption patterns, lack of adequate public awareness concerning the deterioration of environment and dilapidation of resources and finally persisting human egoism make both the protection of the environment as well as the quality of life difficult goals to reach.

The increasing concentration of human populations in cities presents urban planners with enormous challenges in meeting the demand for infrastructure. Population growth and high human densities can exact a heavy toll on a city's fragile natural and environmental resources, particularly in developing countries where squatter communities are prevalent and resources are scarce. Nevertheless, the preservation of a system of vegetated areas, or green spaces, can improve the quality of life by providing people with natural settings for leisure and recreation, and by safeguarding the quality of precious life-giving resources such as air and water. Green areas also have the potential for affording citizens the opportunity to get direct economic benefits through urban agriculture or forestry. Yet none of these amenities occurs accidentally.

A building's thermal performance and energy consumption is influenced by its layout, external shading devices, the surface colour, and insulation. Highly reflective finishes for walls, openings, and roofs, which provide a high amount of albedo, can also indirectly improve a building's thermal performance (Misni & Allan, 2010). However, proper landscaping, water features, and other landscape structures can also be used to improve thermal performance. They reduce the amount of radiation falling on the building through the provision of shading, and by moderating temperatures, the evapotranspiration processes,

and controlling wind direction to assist in keeping the building warm or cool. The design of external spaces needs to be prioritised because it will influence the building interior.

Urban parks and other vegetated areas in cities have traditionally been viewed principally as recreational amenities. Urban greening comes from the recognition that such urban green areas can and should be used in an integrated, holistic manner for many other environmental and social benefits beyond recreational use and aesthetics. These include improving basic sanitation, providing potable water, controlling floodwaters, treating sewage, reducing air pollution, disposing of solid wastes, moderating both macro- and microclimates, increasing bio diversity and reducing poverty.

1.1 BACKGROUND STUDY

Just like water, electricity, education and health, food are considered essential public services, the role of urban green spaces is equally important in establishing and maintaining quality of life in a community. They contribute to health and environmental wellbeing of a community and the region. The physical elements and activities in the urban green spaces offer many benefits to the citizens, namely health, social interaction and economic value. They provide identity and sense of place to the citizens.

Urbanization results in an increasing proportion of the population living in cities. Urban living limits access to nature and can increase exposure to certain environmental hazards, such as air and noise pollution. Many urban areas face increasing pressure from expanding populations, limited resources and growing impacts of climate change. These challenges must be addressed in order for cities to provide healthy and sustainable living environments.

The green spaces are the most significant social spaces for people to interact on a daily basis and also considered as one of the sustainability indicators for maintaining the well-being in residential Communities. The benefits of green space for wellbeing are extensively recognized and progressively more documented. Due to increasing urbanization and housing demand, Residential communities are growing in suburbs and few in the urban core. Due to which depletion in per capita green spaces are recorded. Much of this work focus on the potential of natural environments to provide opportunities to enhance wellbeing in different ways:

Physical wellbeing through physical activity and fresh air.

Mental wellbeing through stress reduction and attention restoration.

Social wellbeing through social integration, engagement and participation.

As more people leave rural environments for the steel and concrete surroundings of cities, there is a growing recognition that they will need and want some form of vegetation present in their daily lives. Whether it is a shady city park for recreation, a tree border for noise reduction or a wetland area for flood control, the concept of urban greening is fast becoming a reality. The term evolved from Millers (1988) definition to mean A an integrated, city wide approach to the planting, care and management of all vegetation in a city to secure multiple environmental and social benefits for urban dwellers. While A urban forestry typically refers to the planting and maintenance of groups of trees and A urban agriculture to food grown by city and suburban dwellers, A urban greening is a more general term.

Urban parks and other vegetated areas in cities have traditionally been viewed principally as recreational amenities. Urban greening comes from the recognition that such urban green areas can and should be used in an integrated, holistic manner for many other environmental and social benefits beyond recreational use and aesthetics. These include improving basic sanitation, providing potable water, controlling floodwaters, treating sewage, reducing air pollution, disposing of solid wastes, moderating both macro- and microclimates, increasing bio diversity and reducing poverty.

This report examines the role of urban greening in Central and South America (hereafter referred to collectively as A Latin America @) and the Caribbean. The report is intended to be of interest to a broad readership, ranging from government officials and city planners to local business owners, concerned community members, and the staff of organizations such as the Inter-American Development Bank. While the objective is not to be exhaustive on any one part of the topic, the goal is to cover the major elements 2 necessary in project initiation. The report seeks to outline the fundamentals of urban greening programs C including the benefits, challenges, and principal elements of successful projects C for a diverse audience of city planners, government officials, and concerned citizens.

1.2.The Urban Environment

The accelerating rate of urbanization is exacerbating the serious environmental problems already found in the cities of Latin America and the Caribbean. The urban poor, usually residing on marginal, environmentally sensitive land, often face the greatest exposure to urban environmental hazards. Despite the fact that the poor are most often the hardest hit by these hazards, air, water and noise pollution affect people of every economic stratum. The social and ecological costs of urban pollution will continue to affect the growing urban populations in the region unless the cycle of degradation and poverty can be broken. Cases of soil, air and water pollution as well as other environmental concerns, can be found in any city in Latin America or the Caribbean.

Water pollution in the region is particularly prevalent. For instance, it is common for industries in many cities to discharge untreated waste water into municipal sewers or to dump it directly into urban rivers. Another source of contaminants is human excrement, which is often deposited in or near a city's waterways. The infectious and parasitic diseases thus spread are among the leading causes of morbidity and mortality in the region.

Air pollution is another major cause for concern. The increasing number of vehicles with untreated exhausts, expanding industrial sectors, and poor natural ventilation in many cities have combined to create serious smog problems. In addition, the use of coal and highly toxic fuels for cooking and heating have helped create unbreathable air in many of the region's cities. This, in turn, has resulted in pollution-induced hypertension, respiratory diseases and other serious health problems.

Urban expansion has also been detrimental to the land in and around cities. Conversion of open space and agricultural lands to concentrated urban development decreases water permeable areas, upsets natural drainage patterns and has caused serious flood problems in many of the region's cities.

Vegetation includes all the plant life in a particular region, such as trees, shrubs, grasses and lawns. Vegetation can influence solar radiation, air temperature, humidity and air flow. While there are a number of landscape strategies that can be implemented to modify the microclimate, the three main functions of vegetation used to do this are: shade, evapotranspiration, and wind control

1.3. Need of urban green space in present world

It is a common saying that human culture began with the first tree and will end with last one. Several studies suggest that growing population and process of urbanization are major factors in the transformation of relaxing green spaces to solid rigid concrete planes. There is no doubt that the trees and forests have always played a significant role in determining the shape of human evolution and development of our cultures and societies, their importance in improving the ecology of cities. This calls for the large role of trees, more of forest and better conducive environment for their growth and management. In the third world countries, the growth and expansion of urbanisation depend largely on cultural, politics and history of regions. Similarly changes in the nature and characteristics of urban settlements depend on individual culture (Shyam Sundar, 1985 and FAO, 1995). Urban and peri-urban areas are increasingly getting CHAPTER-2 LITERATURE REVIEW Nartan Rajpriya, Department of Geography, Gujarat University 25 exposed to ill effects of urbanization. Urban fringe areas are called as transition zone, where modification occurs rapidly due to the interaction of rural as well as urban human activities. Moreover, majority of the urban population resides in fringe zones and therefore their living conditions may be threatened by the frequent changes of environmental dictators which again occurred by human activities. (FAO, 1995) The peri-urban areas are also subjected to land cover change, leading to transformations in the hydrological, ecological, geomorphologic and socio-economical systems. Incidentally, these areas are often neglected by both rural and urban administration, particularly from the ecological and environmental consideration. The article gives a thought, that there is a strong need to conserve the existing green areas and to rethink about the urban planning in developing countries. This is more crucial because these countries mostly have high population with thick population density, which is the main issue of lesser amount of land to create new green areas.

1.4. Concept of Urban Green Spaces:

The world's cities are becoming increasingly congested and polluted. Urban green spaces provide a wide range of ecosystem services that could help combat many urban ills and improve life for city dwellers.

The definition of urban green spaces which is agreed on by ecologists, economists, social scientists and planners: "they are public and private open spaces in urban areas, primarily covered by vegetation which is directly or indirectly available for users (Haq, 2011). In

another way, it is the land that consists predominantly of unsealed, permeable, “soft” surfaces such as soil, grass, shrubs and trees. It is the umbrella term for all such areas whether or not they are publicly accessible or publicly managed. It includes all areas of parks, play areas and other green spaces specifically intended for recreational use, as well as other green spaces with other origins

Urban Greens are green spaces located in urban areas mainly covered by vegetation, which are directly used for active or passive recreation, or indirectly used by virtue of their positive influence on the urban environment, accessible to citizens, serving the diverse needs of citizens and thus enhancing the quality of life in cities or urban regions.

Whereas, in context of the United Kingdom planning documents, Green spaces are an area of grass, trees, or other vegetation set apart for recreational or aesthetic purposes in an otherwise urban environment. Green spaces are the ‘green lungs’ of our towns and cities which contribute to improving people’s physical and mental health by providing places for informal recreation – walking, cycling, sitting, socializing and children’s play – and ‘breathing spaces’ to take time out from the stresses of modern life. They include not only areas to which the public have physical access, but also visual access, for example, in the way green spaces provide setting for building, communities, and everyday activities.

Yet in the Indian context, green spaces of the residential areas are termed as planned open spaces, which come under the local open space system. Detailed layout of residential sectors will have to give due consideration to the provision of formal and organized second level open space corresponding to the local open spaces forming an integral part of the built form. Local open space system is envisaged to be linearly structured establishing continuity engulfing the recreational and community based social facilities, the third level of open space corresponds to system of open spaces provisions made within sectors and neighbourhoods consisting of neighbourhood parks linearly inter-linked with an overall open space network.

The above green spaces definitions clearly state that in the European context, the green space definition includes the function, user category, accessibility and its impact on quality of life. Whereas the definition in context of the United Kingdom elaborates green spaces in terms of its physical characteristics, function performed, perceived benefits availed and about physical and visual accessibility. However, in the Indian context green space is

considered for provision in the terms of accessibility and connectivity. According to above discussion the green space might be understood as.

Green spaces are those planned open spatial layouts with visual and physical.

Accessibility in a residential community, which are committed to guarantee services including passive and active recreation. As an impact interface they can enhance social cohesion, feelings of safety–security as also other perceived benefits to all users.

1.5. Urban Planning

In order to maximize the value of urban greening projects and minimize the associated costs, city planners need to be thinking about greening possibilities whenever they are considering urban public works projects. Retrofitting a paved or otherwise developed section of a city with a park or greenway is generally far more expensive and controversial than incorporating greening as part of a project in a previously undeveloped area. Quantitative Targets : When designing a city's green space infrastructure, planners should aim to fulfil certain quantitative targets. The World Health Organization (WHO) recommends that cities provide 9m² of undeveloped (unpaved) open space for every inhabitant. The WHO also suggests designing green area networks so that all residents live within a 15 minute walk of an open space. Other proposed goals are: the development of a biodiversity index used to rank green spaces based on the number and percent composition of native species; maintenance of a permanent record of changes in air and water quality; an increase in cost-effectiveness from project to project; and charting public use and appreciation by measuring user-days (equal to the number of visits per person per year summed over all users), willingness to pay, or other preference indicators. Maximizing Use and Benefits : Planners are also needed to design urban green areas in a way that maximizes the potential uses. Whereas urban green areas have been traditionally designed for recreation and aesthetic value, their usefulness far exceeds these functions. With proper design, green areas can also improve air and water quality, protect biodiversity, reduce erosion and flood risks, provide agricultural output, etc. But all this requires planning. It requires cross-sectoral communication where city planners talk with water and sewage

engineers, transportation specialists, agronomists, businesses, and local communities to design green areas that best serve the intended beneficiaries. There are myriad possibilities of implementing green areas outside of the traditional contexts of city parks, tree-lined streets, or greenways. For example, creative planners have integrated green areas into water treatment projects, business parks, highway construction, flood-plain protection, and urban farms. One of the additional benefits of integrating green areas into other projects is that some or all of the costs are absorbed into the budgets of these projects. Greening costs become a natural part of infrastructure programs when developers are required to set aside project areas for ecological benefits, or when urban planners work in conjunction with the transportation or water and power departments. Lastly, planners must think ahead about ensuring the longevity of the green spaces they create. It is not enough to secure the funds for the establishment of a particular green area. That area also requires funds for maintenance, protection and monitoring. Without this money and public participation, the plants die, the soil is compacted, the area may be vandalized or covered with litter, and could become crime-ridden or taken over by squatters, thus losing the benefits that they were designed to generate. The best chance for a green area to remain a viable source of multiple amenities to surrounding communities comes not only when there are provisions made for upkeep, but when there is citizen participation. Because a safe, well-maintained green area provides multiple benefits for the surrounding community, planners should be able to stimulate citizen and local business participation in the upkeep of the area. By garnering local participation, the community is made a vital and effective stakeholder. Once a community feels a sense of ownership of its green space, the citizens will help provide far more effective maintenance and protection than the city alone could provide.

1.6. Green spaces in cities

Green spaces have had a part in urban planning for many centuries (FloresXolocotzi, R., et al., 2010). Nevertheless, their relationship has been changing, in continuous evolution, following different trends and tendencies that have occurred through the years (Groenewegen et al., 2012; Maheswaran, 2010; Jansson, 2014; James et al., 2009). Therefore, these spaces also could be used to express contemporary values, cultural trends, tendencies, and beliefs of urban societies (James et al., 2009; Jansson, 2014). Green spaces

represent both the culture in which they were designed and the communities that make use of them, since different populations value differently the contact with nature. This means, the role these areas have differ from one country to another, and can also differ within the city, in regards of different social groups (James et al., 2009).

The differences among countries in the EU can also be explained by other factors, such as the climate or geography they have. For example, it has been proved that green spaces are less numerous in southern cities, where climate and vegetation usually form landscapes with few green expressions (Fuller et al., 2009). This also means that in these cities, the maintenance of green spaces will carry more costs than cities located in areas with more humid climates, for example. The amount and quality of green spaces therefore tends to improve in European countries located towards the north and northwest of the region (Fuller et al., 2009). The management of these areas therefore varies widely across regions, but also within the same country (James et al., 2009). Several disciplines are involved in the planning, design and management of resources to develop these areas (James et al., 2009), and as some experts point out, this coordination of disciplines is of high importance (James et al., 2009). In addition, it is also important to highlight the fact that, even though nowadays the plans and measures towards a more 'green', sustainable city are already developed in almost every city, and moreover, every district in a city have a different background (Stockholm City Council, 2010). This is, as the city has been growing, new developments have been designed and added to it. Some neighbourhoods or districts were planned as a result of a demand, which means that some areas of the city are occupied mainly by one land use, such as housing, or office buildings. This fact hinders the development of green spaces in these areas, since not only they need to develop new areas that fulfil the new standards, but they also sometimes have to change the already present part in order to make a more cohesive city. Thus, it is complicated to establish a unique management guideline for green spaces, since they are required to fulfil different roles and functions depending on the area or region (James et al., 2009). Usually, the management of these areas rely on a municipal or regional level (James et al., 2009). For this reason, the focus of this thesis will be put in the two capital cities of both countries, so as they have a similar administrative role and importance.

Nevertheless, in recent years where urbanization processes have increased and are expected to continue to in the recent future, the need for space in cities have endangered the

presence of these areas (Andersson et al., 2014; de la Barrera et al., 2015; Groenewegen et al., 2012; James et al., 2009; Maas et al., 2006; Maheswaran, 2010). Europe is a crowded continent (Fuller et al., 2009), demand of space for housing, public infrastructure, etc. have evolved into a lack of development of new green spaces in the cities, decreasing therefore the access some parts of the population have to them, as the boundaries (both physical and political) for most cities area already designed (James et al., 2009; Jansson, 2014), and impossible to change.

This issue has deteriorated with planning measures that aim for the densification of the city, in order to combat the effects of spread out populations (Jansson, 2014; Maas et al., 2006). These policies have resulted in the decrease the green spaces in the process of creating a dense urban scene. Populations therefore live in urban spaces with fewer green spaces (Fuller et al., 2009; Jansson, 2014; James et al., 2009; Maas et al., 2006). In some cases, new, largegreen spaces have been built but in the outskirts of the cities, where there's a wide availability of space, and the costs of land are not as high.

Nowadays there are multiple factors and developments that are forcing experts to re-examine the way a city is designed, planned and experienced (James et al., 2009). In their article, James et al. (2009) highlight that, despite the fact that the functions these green spaces fulfil are becoming more well-known now than ever (also supported by Jansson, 2014), there's still the need to redesign the integration in planning of these areas, in order to maximize their services and benefits (Andersson et al., 2014), as well as the need to design new approaches to support the decision-making process related to these areas (James et al., 2009), and therefore gain a more prominent role in spatial 20 planning practices (Jansson, 2014; de Vries et al., 2003), so that the role they provide is not undervalued (Jansson, 2014). Maintaining both quantity and quality of these green spaces should be a priority issue in planning practices globally (Fuller et al., 2009; Jansson, 2014), so that the services provided by these areas can not only be maintained, but even increased

1.7. CHARACTERISTICS OF URBAN GREEN SPACES:

Such green spaces vary in size, vegetation cover, species richness, environmental quality, proximity to public transport, facilities and services. Public green spaces include parks and reserves, sporting fields, riparian areas like stream and river banks, greenways and trails, community gardens, street trees and nature conservation. Public green spaces should be at the centre of neighbourhood and not more than five minutes' walk for most residents, public buildings or shops. Therefore, accessibility and proximity are very important factors to consider during planning and design of an urban green space.

In green space research, there is an increasing number of studies that concern people's

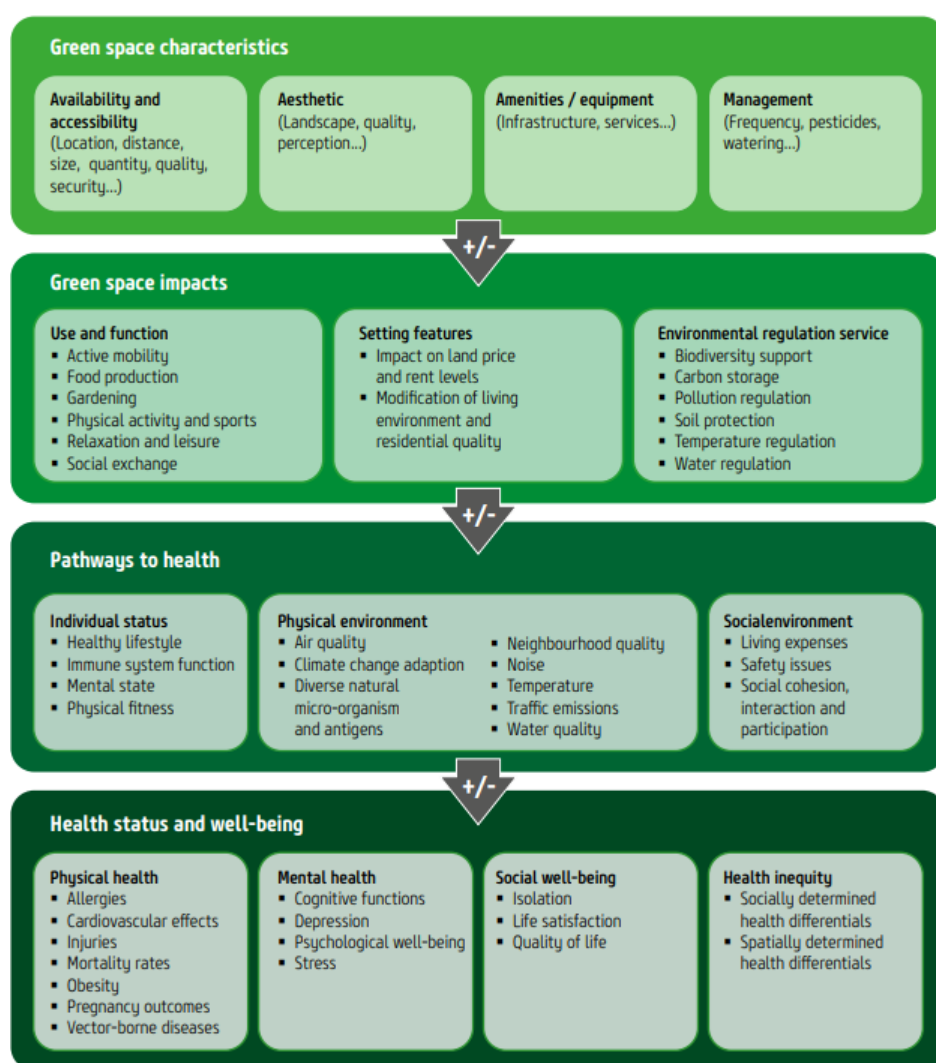


Figure 1: 1.7. CHARACTERISTICS OF URBAN GREEN SPACES

experience and use, but it has been pointed out that more research is needed due to increasing urbanization issues like safety and security. Many previous models and theories substantiate that the usability of green space is directly associated with its accessibility. The usability of

green space differs due to different spatial arrangements within the residential communities and variation in the provision of it with respect to quantity and quality. Physical accessibility and the connection is a major research area in the field of green space use in the residential area because people are associated with the green space usage

1.8. IMPORTANCE OF GREEN ENVIRONMENTS FOR RESIDENTIAL COMMUNITIES:

In many countries like England, Australia, Germany and Belgium the concept of the green environment is popular they explored healthy city planning and defined it as an effective tool for achieving sustainable health and well-being in humans. It is important that these types of green environments are publicly accessible to provide spaces of social interaction for residents. The green spaces have diverse classifications, and each provides its own values and benefits. Thus, further exploration of green environments is critical to ensure higher accessibility, usability, and functionality of green environments for future residents. In many residential communities green spaces are buffers between Built and Exterior surroundings. In addition, Multi unit residential developments have focused on branding value and esthetic aspects and ignored the importance of natural spaces within the living environment.

However, the importance of green spaces in residential communities has been recognized by residents and in some communities they are developed and protected voluntarily. Hence it is essential to understand the importance of green spaces for sustenance of the well being of residential communities and their overall development.

1.9. GREEN SPACE FUNCTIONS AND BENEFITS:

1.9.1. General benefits:

The functions and benefits of green space are plenty in number. Green Space symbolize peace, minimal stress and a cleaner environment for many people, which are considered as important factors in making a city livable, pleasant and attractive for its citizens. The primary functions performed by Urban Green Space are given in Table below.

S No.	Type	Functions
1	Regulation functions	Regulates the chemical composition of atmosphere and purifying the local air
		Controls the runoff and flooding
		Regulates the hydrological cycles
		Supports biological diversity in the city
		Prevents the soil erosion and sediments
		Regulates the local and global climate
2	Carrier functions	Conserves the energy in the city through control of the micro climatic variations
		Help recreation and tourism
		Integrates urban man to nature.
3	Production function	Recharging the ground water table
		Providing medical resources
		Providing Raw material for some of the human activity
4	Information function/ dissemination	Esthetic information
		Spiritual and religious information
		Cultural and artistic inspiration
		Scientific and educational information sources

Figure 2: GREEN SPACE TYPES AND FUNCTIONS

1.9.2. Benefits of Green spaces: The Ecosystem Services Perspective

Recent studies have been developed in relation with green spaces, that seek to understand the positive effects green spaces have on both the population around them and the city as a whole. The effects these areas provide are numerous, and relate to a wide range of different aspects of the city. The effects depend on the landscape qualities of the green space, such as the size, the shape, the placement and the content of the area (de la Barrera et al., 2015; Jansson, 2014). They affect the city differently, at several levels at the same time.

In order to highlight these benefits, the 'ecosystem services' approach will be used (Andersson et al., 2014; Niemelä et al., 2010; de la Barrera et al., 2015; Jansson, 2014). This approach is used to highlight the benefits green spaces have in relation with the population, as well as categorize them in three main groups, as it will be explained later. As Niemelä et al. pointed out, (2010), this approach can help to eliminate the existing gap between urban natural and social scientific research, so that the academic findings can be successfully applied in urban planning and governance and in the maintenance of biological diversity (Niemelä et al., 2010). Even though, as mentioned before, benefits of green spaces are being researched, and in some cases applied to policies, ecosystem services in cities have not been studied intensely (Niemelä et al., 2010; de la Barrera et al., 2015). This situation has lead towards a state in which 'land-use planning and management decisions are based on inadequate information on the benefits that humans can derive from these areas' (Niemelä et al., 2010, page 3227). This lack of information can cause the underestimation of the ecosystem services provided by these areas, that might lead to an unawareness of the impact that changes in land-use can cause to these services (de la Barrera et al., 2015; Niemelä et al., 2010). If these impacts are not addressed in time, they can lead to costs caused by the need to restore or reestablishment of these services (Niemelä et al., 2010). Part of these costs can be assessed in economic terms, but other costs may be difficult to quantify (Niemelä et al., 2010; de la Barrera et al., 2015). As Andersson et al. (2014) pointed out, 'incentives, interests and ambitions among managers and stakeholders and institutional framework set the stage for management of such spaces and their ecosystem services' (Andersson et al., 2014 p 446). The ecosystem services approach could help to develop a comprehensive understanding of this concept that reaches civil servants, decision makers and even inhabitants (Niemelä et al., 2012).

This concept offers a more inclusive approach to the processes undergoing through the whole city, emphasizing their influence and relation (Andersson et al., 2014; Niemelä et al., 2012). This perspective also offers the possibility to address nature issues from an economic and service production perspective, a focus that, even though academically doesn't present many benefits, could help to stress the importance of these services to government representatives, private companies and such. This approach also offers a way to develop ecologically sustainable urban regions, since it brings together the functioning of humans and society, as well as human activities. This is, 'ecosystem services are a functional link between humans and nature' (Niemelä et al., 2012, p. 3238). These services have a different range of impact, as some of them may have results at for example municipal level (for example, an increase in the air quality due to the present of trees within the city), but some of them may be effective only locally (for 12 example, the decrease of temperature due to the presence of trees that provide shadow in an area of a neighbourhood). (Niemelä et al., 2010). In addition to this, the effect of some of the services depends upon the quality and quantity of the green space in which they are represented, connectivity among spaces, species diversity and composition in these spaces, population densities, etc (Niemelä et al., 2012).

The benefits green spaces offer have been classified following this distinction of ecosystem services:

I. Provisioning services

The provisioning services, that include aspects such as food production (an activity that is increasingly popular in some European cities in recent times), water etc, this is, material benefits that the ecosystem provides as Niemelä et al., mentioned (2010), won't be included in this thesis, since it is focused on urban environments, where the existence of these services is very limited.

II. Regulating services

The regulating services is the services that regulate the ecosystem, and formulate the essential background for the development of other ecosystem services (Niemelä et al., 2010). These regulating services can be seen in for example the presence of green spaces that has helped to improve the 'health' of the city (Hartig et al., 2014; Jansson, 2014; James et al., 2009; Fuller et al., 2009). These green spaces, along with the vegetation that usually

accompanies them, can help reduce noise levels in the city, and depending on the vegetation used, it can absorb or soften noise generated from different sources, such as traffic or the development of other activities within the city, such as industries (Jansson, 2014; Niemelä et al., 2010).

The presence of this vegetation can also improve air quality, as the plants will filter the air and reduce the amount of certain pollutants present in it (Hartig et al., 2014; James et al., 2009; Jansson, 2014; Niemelä et al., 2010). Vegetation can also help to reduce the urban heat island effect that most cities suffer from, i.e. a rise of temperature in comparison to their surrounding areas, as a result of the development and concentration of human activities in the area. Vegetation can reduce heat islands by providing shadow, and therefore also reduce energy demand (Hartig et al., 2014; James et al., 2009; Jansson, 2014; Niemelä et al., 2012). Green spaces, especially those with a certain dimension, help to ease the issue also by providing a building free space in which air currents can circulate and consequently alleviate the heat in this areas (Jansson, 2014).

The impact on these aspects depends on the constitution of the green spaces, since their effects vary depending on the vegetation density, the vegetation species used, etc. (Hartig et al., 2014; Jansson, 2014). As Jansson (2014) mentions, the differences in temperature generated by a green space can vary between a reduction of 1 - 4 °C. Some authors argue that green spaces help to ease the negative impacts urbanisation has in both the population and the environment (Maheswaran, 2010; Niemelä et al., 2010). In addition, It could also be said that the presence of these areas can determine the sustainability of an urban development

III. Cultural services

The cultural services are those immaterial benefits that citizens can obtain from the ecosystem (Niemelä et al., 2010). Cultural ecosystem services are considered to be the most important ones for city inhabitants (Niemelä et al., 2012). For example, green spaces also have an effect on population. These areas are places that can also improve social cohesion,

by providing a safe and close space for neighbours to meet, increasing therefore the sense of community, as well as the sense of safety, of wellbeing, of happiness, etc. (Hartig et al., 2014; Groenewegen et al., 2012; Jansson, 2014; James et al., 2009; Maheswaran, 2010; Fuller et al., 2009), while also creating more cohesive communities (Groenewegen et al., 2012).

This effect on population is deeply linked with the sense of security perceived by the individuals, as if the area is not perceived as safe, social interactions may be reduced or even not produced at all (Maheswaran, 2010).

Therefore, green spaces must be maintained, monitored and offer varied facilities, improving the sense of security they produce, so they can reach their full potential and increase the benefits they provide (Hartig et al., 2014; Jansson, 2014; Maheswaran, 2010; Niemelä et al., 2010), otherwise they might have a negative effect on people (James et al., 2009), such as a decrease in perceived safety and therefore experience a decrease in their use, or the emergence of animal pests that can compromise the health of the people using them (James et al., 2009). The maintenance of these areas can be quite challenging, especially in growing urban regions (Niemelä et al., 2010), or in regions that don't have enough resources to face these investments.

It has been proved that green spaces have also beneficial effects on the health of the population. For example, the presence of these spaces can encourage the development of open-air activities, such as going for a walk or running, therefore improving physical activity and decreasing the effects certain 'modern' illnesses have on individuals, such as obesity by providing spaces in which these activities can be conducted. This issue is especially relevant, since nowadays there is a high increase of working positions computer-related that are increasingly driving the population to sedentary lifestyle (Niemelä et al., 2010). This effect on population is also connected to the sense of security perceived by the individuals, as if these spaces are not perceived as safe, their use will decrease considerably, affecting special certain population groups with reduced mobility, such as children, disabled or elderly population

The quantity of green spaces refers to the relation between the area of green spaces and the total amount of population, and the urban context (de la Barrera et al., 2015). The quality of green spaces is often related to the amount and state of vegetation present in these spaces, measured with the density of the coverage of vegetation in these areas (de la

Barrera et al., 2015). However, there are several other factors that could be used to study the quality of these spaces. Accessibility is studied by the spatial distribution of these spaces within the urban area (de la Barrera et al., 2015).

On the other hand, these three elements represent a duality that hinders the future planning and developing of these areas. In one hand, in small cities new public green spaces should be created next to existing ones, so that publicly owned green spaces (for example, parks) and privately owned ones (for example, private gardens) support and complement each other. And on the other hand, it is highly important to place these new green spaces in places that are lacking them (Niemelä et al., 2012). It is therefore highly important to be aware of the ecosystem services urban regions provide, and how human actions shape them (Niemelä et al., 2012). These benefits obtained with almost daily contact with green spaces are accumulative (Hartig et al., 2014; James et al., 2009), and influenced by various aspects at the same time (Maheswaran, 2010), which means that individuals who have access to them on a daily basis will have a better self-reported health than those who don't experience these places so often.



Figure 4: Benefits of Green spaces



Figure 3: Healthier Community

1.9.3. What do Good Urban Green Spaces Offer?

Urban green spaces can give enormous benefits to the neighborhood and community by improving health, social well-being and enhancing enjoyment of local environment. The important role of urban green spaces is recognized both in the character and the life they bring to towns and cities around the world (Malek, Mariapan and Shariff, Unknown).

1. Environmental Benefits

Ecological benefits: urban green spaces supply cities with ecosystem services ranging from maintenance of biodiversity to the regulation of urban climate.

Pollution control: green space may filter air, remove pollution and attenuate noise.

Biodiversity and nature conservation: green spaces do functions as protection center for reproduction of species and conservation of plants, soil and water quality. Urban green spaces provide visual relief, seasonal change and link with the natural world.

2. Economic and Aesthetic Benefits

Energy savings: using vegetation to reduce the energy costs of cooling buildings has been increasingly recognized as a cost-effective reason for increasing green space and tree planting. Plants improve air circulation and provide shade. This provides a cooling effect and help to lower air temperature.

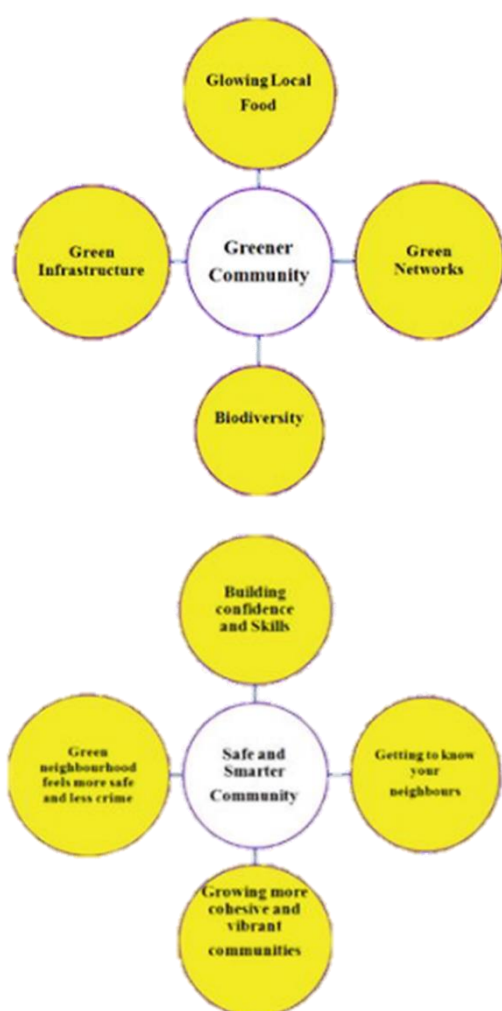
Property value: areas of the city with enough greenery are aesthetically pleasing and attractive to residents, visitors and investors. Urban green spaces can be one of the factors that attract significant foreign investments that assist in rapid economic growth (Haq, 2011).

3. Social and Psychological Benefits

Recreation and wellbeing: green spaces are key element for recreation and to experience nature. They enhance the quality of life for people living in the surrounding area and provide locations for a variety of leisure activities as a multi-use recreational open space. Green spaces are concerned with social values and how places encourage people to interact in ways which lead to trust, mutual understanding, shared values and supportive behavior. Social values arise when people can connect to others with common interests.

Human health: help in improving the health of the public, as it is discovered that time spent in nature, relieves mental fatigue and the feelings of violence and aggression that can spring

from it. Natural resources provide many “activities” that require little to no effort, yet provide ways to restore a person’s health and mental wellbeing. This kind of green spaces can help to relax and unwind and create social development in children. Also, physical health and wellbeing can be maintained through participating in organized sports, exercising informally, contact with nature and being outside in the fresh air and sunlight. Recent scientific studies have demonstrated that natural areas have positive health impacts on development issues, particularly behavioral disorders like attention deficit hyperactivity disorder (ADHD). Health studies have also shown that contact with nature offers a range of medical benefits such as lower blood pressure and cholesterol levels, enhance survival after a heart attack, a more rapid recovery after surgery and lower levels of stress and depression.



The recent model for the benefits of urban green space as shown in Figure suggested by Bedimo-rung et al. [8] describes the relationships between park benefits, park use, and physical activity, and the antecedents/correlates of park use. In this classification scheme, the discussion focuses on park environmental characteristics that could be related to physical activity, including park features, condition, access, esthetics, safety, and policies which is responsible for well being of residents.

Urban Greening benefits:

Urban greening offers improvements in air, water, and land resources by absorbing air pollutants, increasing water catchment and floodplain surfaces, and

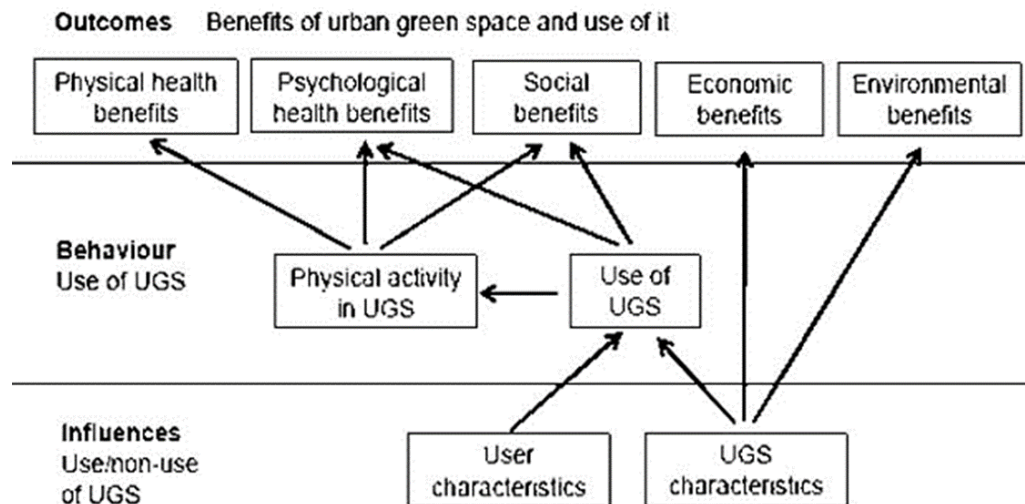


Figure 5: Benefits of Urban Green spaces

stabilizing soils. Urban forests act as temperature buffers C providing shade in the summer, and wind break in the winter C in addition to reducing noise pollution and CO₂ levels, and providing a habitat for wildlife. The economic benefits include marketable timber and agricultural products, as well as a variety of nontimber forest products such as artisan wares and honey. Lastly, the overall benefits to society, particularly to low-income residents, are significant. They include the contribution of trees and vegetation to the mental and physical health of the populace, and the provision of recreational opportunities and an outdoor classroom for environmental education. In addition, they provide aesthetic improvements to an environment otherwise dominated by asphalt and concrete.

Environmental Benefits

- ❖ Air quality improvement
- ❖ Climate improvement
- ❖ Energy saving
- ❖ Solid Waste and Land Reclamation
- ❖ Noise Abatement
- ❖ Waste water treatment
- ❖ Erosion control

Air Quality Improvement

While air pollution indices in many cities in more developed countries have dropped over the last ten to twenty-five years, air pollution levels have been rising in cities throughout much of Latin America and the Caribbean. In Mexico City, for example, the average level of particulate suspension in the atmosphere rose from 65 mg/m³ to 400mg/m³ between 1974 and 1990 (Carter 1993). One study rated the air contamination of Mexico City as the eighth worst in the world, surpassed only by levels in Beijing, Calcutta, Delhi, Jakarta, Milan, Shanghai and Tehran. Suspended particulate matter, carbon dioxide and ozone are also serious problems in Sao Paulo (Carter 1993). Those most affected by such detrimental air contaminants are children, the elderly and people with respiratory problems. Topographical features frequently exacerbate the hazards of air pollution in certain cities. The mountains that ring Mexico City and Santiago, Chile, cause inversions and other climatological phenomena that worsen air pollution problems. Therefore, in these cities an aggressive and multifaceted approach to combating pollution is all the more urgent. Using vegetation to reduce air pollution is an effective technique that also provides other benefits such as city beautification.

Urban greening can reduce air pollutants to varying degrees. Air pollution is directly reduced when dust and smoke particles are trapped by the vegetation. In addition, plants absorb toxic gases, especially those from vehicle exhausts, which are a major component of urban smog (Nowak et al. 1996). Santiago, Chile, which has a serious air quality problem, has recently named its urban parks program the A Lungs of Santiago to reflect the benefit of urban vegetation in reducing the city's air pollution.

Climate Improvement

One of the most important benefits of urban vegetation is its impact on the climate. Two distinct influences can be identified. First, there is a direct effect on human comfort. Secondly, there is an effect on the energy budget of buildings in cities where air-conditioning is used. Both of these effects can be significant, or negligible,

depending on the size, spacing and design of vegetated areas. The direct impact on human comfort is one that every person is familiar with, although it is hard to quantify. Anyone who has walked on a city street on a rainy, hot, or windy day knows from personal experience that trees can significantly increase human comfort by influencing the degree of solar radiation, air movement, humidity and air temperature and providing protection from heavy rains.

Wind speeds 2 meters above the ground in a residential neighbourhood were shown to decrease by 60 percent or more in areas of moderate tree cover compared to open areas (Heisler 1990). The urban poor in Latin America and the Caribbean have a further appreciation for trees for the shelter they can provide in lieu of a more permanent structure. Trees and other vegetation can also have an important impact on the energy budgets of buildings and, in turn, of entire cities.

This effect is most noticeable in urban centres with little or no vegetation and extensive paved areas. Large areas of paved surfaces dissipate the heat of the sun only very slowly. This results in the urban heat island effect where a city heats up rapidly and then maintains a high temperature. Furthermore, as city temperatures increase, so too do airborne pollutants and smog (Kuchelmeister 1991). Studies in Sao Paulo show that the Integrated Air Quality Improvement Mexico City emits 4,300,000 tons of contaminants into the air annually.

Its geographic location in a valley surrounded by mountains exacerbates air pollution by reducing dispersion of the emissions. Warm winds from the northeast create thermal inversions and increase pollution levels further by trapping contaminants in the valley. In addition, because the city is 2,200 meters above sea level, the oxygen content of the city's air is approximately 23 percent lower than it would be if the city were at sea level. Air pollution levels exceed Mexican safety standards approximately 250 days of the year.

In response to this dire situation, Mexico City officials are applying four principal strategies to improve the quality of their air: 1) improving the quality of fuels used in the city; 2) reorganizing the urban transport system; 3) mandating emission control systems; and 4) reforesting the city and suburban areas. The IDB is currently providing partial financing for a US\$200 million Mexico City Environmental Conservation Program with urban and suburban reforestation components (IDB 1992). 10 temperature in the concrete

-covered centre of the city is consistently several degrees warmer than in more open and vegetated outlying areas (See Appendix A, Photo #2). Akbari et al. (1992) found that tree shade could reduce the average air temperature in buildings by as much as five degrees Celsius. Expanded research is currently being conducted in several climates in the United States and Puerto Rico through a joint partnership between Cool Communities (a research program of American Forests) and the U.S. Department of Energy. The U.S. Forest Service is also studying the effects of vegetation on urban climates as diverse as Austin, Texas and South Dade County, Florida. These studies will no doubt be of benefit to city planners in Latin America and the Caribbean and in other developing regions.

Energy Savings

Various components of an urban greening program can contribute substantially to reducing a city's overall energy budget. As mentioned previously, inner city concentrations of pavement and concrete produce an urban heat island effect that causes discomfort and health problems, especially for the poor who cannot afford air-conditioning. On the other hand, cooling air-conditioned buildings requires considerable amounts of energy. High electricity demand during hot months can often cause brown outs. This can be greatly reduced by planting substantial amounts of vegetation in dense downtown areas. To date, the most extensive studies on the influence that vegetation has on the energy budget of cities have been carried out in largely temperate climates in developed countries. Studies in Chicago show that increasing tree cover in the city by 10 percent may reduce the total energy used for heating and cooling by 5 to 10 percent (McPherson et al. 1994). Urban green areas can supply fuelwood and other substitutes for imported fossil fuels, thereby providing sources of renewable energy.

According to Smit (1996), food production amounts to approximately 40 percent of the economies of Latin American cities. Producing it in urban and neighbouring green areas reduces transportation costs to market. Studies done in the United States show that it may take seven or eight calories of fossil fuel for rural agriculture to deliver one calorie of food energy to the consumer. Urban agriculture uses only an average of one calorie of fossil fuel for each food calorie delivered. Another

area where large energy savings can accrue is in waste management. Treating waste waters in biological systems (see discussion above) eliminates the need for major sewage treatment plants that operate with costly usage of fuel. Similarly, handling solid waste at its source in each neighbourhood (using organic wastes for animal feed, composted fertilizer and mulch for green areas), reduces the energy costs associated with trucking it away and treating it in landfills or elsewhere. Given the large populations of many of the region's cities, these savings can be quite substantial.

Water Supply Catchment Protection

Contaminated water sources and waterborne diseases cause a high percentage of illness and death, especially in infants and young children, throughout Latin America and the Caribbean. Typically, the poorest communities are hardest hit by the problems associated with unsanitary water supplies. One of the significant challenges for city planners is providing potable water to all the city's inhabitants. Given the importance of maintaining quality water supplies, it is imperative that cities adequately preserve their water supply catchment areas.

Determining how the costs of protecting such areas will be allocated between urban and rural residents, who both need and use these resources, is a complex issue of equity and practicality. Water catchment areas for most cities tend to be in close proximity to their location and are often found in the suburbs (defined as areas of urban sprawl on the perimeter of urban centers and sometimes referred to as A peri-urban).

Cities in the region have employed a diversity of strategies to protect the suburban or rural areas that serve as the source of their water supplies. Since many of these watersheds are fairly close to the city's edge, some of them have inevitably been depleted or polluted, and those cities now have to pipe water in from quite a distance away.

Wastewater Treatment

The practice of at least partially treating waste water in stabilization ponds has to be considered an alternative for urban waste-water treatment as much as its economic and ecological advantages may indicate. Most cities in the region have significant wastewater treatment challenges and could explore whether

stabilization ponds, integrated into park systems, would economically and ecologically serve some of their treatment needs. The ponds, rivers and wetlands that become part of natural wastewater treatment or pre - treatment can also serve for recreation, wildlife habitat, aesthetics and educational use.

Wet lands are among the worlds most biologically diverse ecosystems. Using wetland resources for tertiary wastewater treatment can significantly increase habitats and enhance the bio diversity of flora and fauna in and along the waterways. For example, a multiple -use park system in Dur ban, South Africa, uses the parks retention ponds and artificial wetlands to partially treat its waste water. Instead of being whisked away through concrete channels (as occurs in many cities), the water entering this park area slowly passes through a series of natural watercourses with floodplains and wetlands. The biological functions and physical aeration that occur in the water during the time it spends in these waterways remove many of the toxic effluents from the wastewater (ICLEI 1995).

These wetlands 12 are utilized as low cost waste water treatment for residents of low -income neighbourhoods close to the park system. In addition, the waterways that allow these natural wastewater cleansing processes to occur also serve the purpose of flood prevention. Wastewaters in the upper parts of the Bogota River watershed are being treated by 22 plants in small municipalities (IDB 1990). Adequately analysed urban runoff waters may be sent to suburban agricultural fields to be used in irrigation, providing needed nutrients for the crops (see Appendix A, Photo 3). Similarly, the University of the West Indies in Kingston, Jamaica, is experimenting with the school s wastewater by sending it through lily ponds to be purified, then applying it as irrigation water on university grounds and in vegetable gar - dens. In both instances, wastewater is essentially recycled into food production, creating a more holistic, integrated system within a citys bioregion. There are several alternatives to wastewater treatment and disposal that can incorporate green areas of various types.

The water can be used to irrigate urban and suburban agriculture and forests, horticultural projects (such as flowers for export), city landscaping and parks, and tree farms. All of these options provide for a safe and productive means of wastewater disposal (Braatz 1993). In arid and semi -arid regions this practice can also recharge

groundwater reserves. For example, wastewater reuse is especially important in Lima, Peru, an extremely arid city.

Since 1969, the city has used stabilization ponds to hold wastewater from some 150,000 residents, returning the treated water to those communities to use for irrigating food crops and fruit trees, to water woodlots and as drinking water for livestock (UNDP 1996). This reuse of city water supplies not only recharges the aquifer, it also reduces the demand on scarce water reserves.

An innovative way to purify wastewater is through the use of duckweed. This prolific plant can completely purify wastewater in 20 days, doubles its weight in two to four days and can yield as much as ten times the amount of protein of soya beans. It has been used in the United States for water purification since 1985 and is sold as a cash crop and as fish feed. Israel exports duckweed as an exotic salad crop to European health food stores. Not only does this weed clean wastewaters that need to be treated anyway, it also is a good cash crop for animal feed or human consumption.

In experiments in Bangladesh, farmers are harvesting about one ton of duck weed per hectare per day and earning more than US\$2,000 a year. Clearly, this is an alter - native worth exploring to treat wastewater (UNDP 1996). When using wastewater for irrigation purposes, city planners need to coordinate with environment officials to avoid potential health risks to farmers and consumers. For example, Santiago, Chile, has had laws regulating sewage irrigation since 1941 but they were not strictly enforced. As a result, when there was a cholera outbreak in 1992, raw contaminated sewage was applied to vegetable crops and the disease was thus spread to consumers. Consequently, the government had to bulldoze several thousand hectares of vegetable crops and relocate farming activities to areas with a safe supply of irrigation water. Such severe measures can be avoided if waste water is properly handled and continually monitored.

Recycling wastewaters into green areas deserves thoughtful consideration by the regions city planners. Recycling wastewater into parks or forested, farmed and degraded lands may also be more economical than finding ways to dispose of it somewhere else. A new computer program offers an exhaustive exploration of wastewater treatment alternatives that can help define the most appropriate system for

a given set of circumstances. The program takes into account the resources and constraints of each particular city and can aid city planners in designing the system best suited to their needs and budgets.

Flood Control

Floods cause considerable damage in Latin America and the Caribbean every year. Buenos Aires suffered over US\$200 million in flood damages in 1985. Estimates of 1993 flood damages from the Mississippi and Missouri Rivers (USA) have been calculated to be in the realm of US\$10 billion. Chile has suffered serious floods throughout the country in the last ten years. In 1988, floods in Rio de Janeiro left 289 people dead, 734 injured, 18,560 homeless, and an estimated US\$945 million in damages (Bernstein 1994). The damage included destruction of roads, bridges, canals, drainage networks, dikes, water and sewage networks, electric power systems, factories, residences and commercial establishments.

In addition, there were the added costs of interrupted economic activity, lost revenues from tourism, and clean-up expenses (in this case, US\$50 million). The flood in Rio left the predominantly low-income population the hardest hit, with limited access to schools, health facilities and basic sanitation for months thereafter. Using wetlands and parks as important components of a city's flood control system is not only recommended but is quite feasible. By locating city parks and green spaces in the flood plains of rivers, streams or other drainage systems, planners can increase the permeable surface area available for catchment, reduce flow rates (compared to non vegetated surfaces such as asphalt), and avoid damage to buildings or settlements that might have otherwise been constructed in the area. This has been the case, for example, in Bogota, Colombia (IDB 1990). Green areas can also mitigate flood damage simply by increasing the permeable surface area in a watershed, thereby decreasing Green Space and Flood Control in Curitiba, Brazil The IguaHu River and a number of its natural feeder springs run through the city of Curitiba. When the city was smaller and less developed, the annual flooding of the river and its tributaries during the rainy season was manageable. With a rapid growth rate and a population of more than 1.6 million, Curitiba started to experience problems: development and encroachment on the

floodplain put businesses and lives at stake. Unfortunately, engineering strategies using diversion and channelling proved ineffective C only transferring the flooding risk to other inhabited areas. City planners decided to try reclaiming the floodplain through the creation of recreational green spaces. In a cross- sectoral effort, zoning laws were changed to limit subdivisions, construction of homes and roads, and to expropriate land for the creation of several parks and artificial lakes. The result is a system of green spaces and low density development that provides effective flood protection and has become the city's principal recreational area. (Talaiye & Biller 1994). 14 runoff rates and abating peak flow levels. Interference with other park uses (such as recreation) only occurs in the short periods that the park's wetlands and flood plains are actually flooded.

Noise Abatement

Noise often reaches consistently unhealthy levels in large cities in Latin America. Poor populations living close to heavy industry, commercial and traffic corridors often get exposed to the highest levels. To make matters worse, the building materials used in low-income neighbourhoods do not insulate the residents from noise as well as the more substantial materials frequently used in wealthier neighbourhoods. Mexico City has a consistent noise level of around 75 decibels (the equivalent of listening to an alarm clock or a police whistle), which often reaches 100 decibels (which can cause ear damage) close to major highways and the city's airport (Carter 1993). Noise pollution levels in Santiago and Rio de Janeiro have also been rated as some of the worst in the region. Trees and vegetation can help reduce noise pollution in five important ways: by sound absorption (sound is transferred to some other object), deflection (the direction of sound is altered), reflection (sound is bounced back to its source), refraction (sound waves bend around an object) and masking (unwanted sound is covered up with more pleasing sounds). Thus, leaves, twigs and branches will absorb sound, as will grasses and other herbaceous plants. Tree or plant barriers will deflect sound away from listeners and, if at right angles to the source, will reflect it back to that source. If the noise passes through and around the vegetation, it is being refracted and thereby dissipated.

Vegetation can also mask sounds to the extent that people will filter out unwanted noise by selectively listening to the sounds of nature (bird songs, leaves rustling,

etc.) over the sounds of the city (Miller 1988). Particularly advantageous to humans is the fact that plants absorb high frequency noises better than lower ones, since higher frequency noises are most distressing to people. The optimum planting design to lessen noise pollution is dense vegetative cover in a range of heights. Such green barriers can be established through out Latin America and the Caribbean along major highways and on the boundaries of noisy industrial and commercial corridors.

Erosion Control

Many large cities in the region are located below, on or above steep hills and mountains, or on coastal slopes. Given the general lack of vegetative cover and the hard seasonal rains most of these cities experience, erosion and landslides have become common occurrences. Informal settlements are hit especially hard by landslides, as they are most often built on marginal slopes. Because of the potential for loss of life and property that eroded embankments represent, many countries enacted legislation long ago to prevent urban development on excessively steep hills. For example, sections of the Brazilian Forest Law of 1959 specifically address this issue. In Bogota, Colombia, an environmental rehabilitation project includes the reforestation of 4,450 hectares in the watershed of the Bogota River, and reducing erosion and sedimentation over about 6,800 hectares (IDB 1990).

The planning authority in Rio de Janeiro estimates that three million people (nearly two-thirds of Rio's favela dwellers) live on the steep hills surrounding the city (Bernstein 1994). Mud slides at the end of the rainy season are a constant threat to people's lives and homes there.

Development on high-risk sites, combined with little vegetative cover, increases susceptibility to the landslides that often follow heavy rains (Bernstein, 1994). Much of that risk can be reduced by planting hardy species to hold eroding soil on steep slopes.

Solid Waste and Land Reclamation

Solid waste disposal in most large cities has become a serious and permanent problem. While some cities have managed to keep pace with their growing

populations, the majority suffer from significant garbage dilemmas, especially in and around informal settlements. Quite often, a substantial percentage of a city's solid waste accumulates in vacant lots and alleys or adorns the slopes of gullies and urban riverbanks. Urban greening can help abate the solid waste disposal dilemma. Many forms of waste and nutrient recycling exist and are already in use in other parts of the world. For example, Asian countries have developed a closed loop @ cycles wherein organic wastes are used to feed chickens, pigs and livestock; settling pond sludge is used for agricultural fertilizer; and wastewaters are recycled into irrigation for aquaculture and agriculture. In Latin America and the Caribbean, a few waste - water recycling systems for irrigation are already operational, notably in Lima, Peru, and Bogota, Colombia. Bogota is constructing systems to treat slaughterhouse wastes and 25 sanitary landfills for the final disposal of solid waste in the metro politan area (IDB 1990). Composting is another viable alternative to waste removal. Organic materials in a city's waste stream can be composted to produce quality soil amendments while at the same time reducing the volume of urban refuse and thus the city's waste disposal expenses.

An example of an effective composting system can be found in Milwaukee, Wisconsin (USA). The city's sewage goes to a special treatment facility that turns it into a soil amendment called Milorganite which is highly valued by gardeners and farmers and is very cost effective for the municipality. A sizeable element in many solid waste disposal sites is tree pruning's and other plant trimmings that can be used as fertilizer or mulch. In poor developing countries, such refuse may have a high value as fuel and fodder.

Wildlife Habitat and Biodiversity

Urban green areas have been found to provide habitats for a surprising number of species and for large populations of birds and animals. Most urban residents are familiar with at least some local species of birds and animals that have adjusted to urban conditions. Where more parks and vegetation exist, local and migrating species can find suitable habitats. In particular, suburban wetlands may offer some of the world's most productive natural ecosystems as transitional areas between terrestrial and aquatic environments (Bernstein 1994).

Wetlands that are incorporated into urban greening projects, including those designed or maintained for flood control and waste water treatment settling ponds, provide particularly important habitats for local and migrating fauna contributing to maintaining a healthy biodiversity in the area. An example of preserving and enhancing wetlands for biological diversity can be found in Bogota, Colombia, where an environmental project is managing irrigation waters and cropping patterns to allow water levels to be restored in the Herrera Lagoon, a wetland of enormous biological wealth and the last vestige of the prehistoric Humboldt Lake (IDB 1990). On a larger scale, urban greening can create or restore biological diversity that will reconnect a city to its surrounding bioregion. Cities are built in an existing ecosystem and often destroy it. The flora and fauna that once lived in that system are either destroyed, displaced or have to adapt to the new urban environment.

This process seriously depletes a region's genetic diversity (both plant and animal) and can threaten with extinction some species that are essential to the area's natural ecosystem and consequently to the resident human population. Incorporating green areas throughout a city can reverse the trend of biological destruction. For example, greenbelts and greenways (see Glossary) can provide biological corridors for wide-ranging species of plants and animals from the surrounding bioregion, allowing them areas large enough to disperse their genetic material, a process crucial to species survival. Urban agriculture can also provide biodiversity on a small but important scale. Diversity protects wild and domestic species from adverse conditions (including natural and economic fluctuations) and thus ensures survival.

1.10. Social Benefits

Health

Although they may be difficult to quantify, the benefits of urban greening to human health can be considerable. Certainly, improvements in air quality due to vegetation have positive impacts on physical health with such obvious benefits as decreases in respiratory illnesses. Perhaps less obvious, however, is the fact that urban forests also reduce stress and improve health by contributing to an aesthetically pleasing and relaxing environment (Nowak et al. 1996). Ulrich (1990) found that convalescing hospital patients recuperated significantly faster when placed in rooms with views of trees and outdoor settings than patients without such views. Urban forests provide a connection between people and their natural environment that would otherwise be missing in a city.

This connection is important for everyday enjoyment, worker productivity and general mental health (Nowak et al. 1996). Shade from trees and the resulting cooler temperatures, especially in the summer, explain why people tend to flock to city parks and green spaces to spend time together during both the weekends and during the week. Thus, they also serve a very important social function. Shade also reduces ultraviolet light exposure thereby lowering the risks of harmful health effects such as skin cancer and cataracts (Heisler et al. 1995). In sum, urban forests supply numerous direct and indirect physical and mental health benefits to a city's population.

Employment

Another important aspect of urban greening is the jobs provided for poor, skilled and unskilled laborers. Urban greening projects are often labour-intensive and provide both initial start-up jobs (soil preparation, planting, etc.) as well as more permanent employment (maintenance, management, etc.). For example a forestry component of Mexico City's urban greening program will require extensive use of unskilled labour. Project managers have estimated that the program will need approximately 3,380 workers to produce and transport plants, 3,700 working in the plantations, 800 in management and more than 100 for protection and surveillance in existing green areas (IDB 1992).

Recreation

Green areas provide recreational sites, especially for lower income residents who tend to frequent city parks more than wealthier citizens because of financial constraints and restrictions on leisure time. This, of course, depends on two conditions: first, the park must be within an affordable traveling distance for the individual or family; and second, it must have the amenities those people prefer. In Latin America and the Caribbean, the preferred activities tend to be barbecuing, playing soccer or volleyball, walking or just enjoying the natural surroundings.

In Mexico City, the centrally located Chapultepec Park draws up to three million visitors a week who enjoy a wide variety of recreational activities (Gonzalez 1996). The urban poor generally have few affordable options for recreation, and thus place a high value on green areas. Nevertheless, their preferred recreational activities may vary from city to city, or even among neighbourhoods. Therefore, research or social surveys about their preferences should be considered as a tool to help city planners design appropriate new green areas. For example, Brasilia was planned more according to central European than Latin American criteria. It does not have the traditional central area, or plaza central, where people can meet to socialize, go for a stroll, or sit in the shade to chat. On the outskirts of Bogota, Colombia, a group of low- to middle-income volunteers cooperatively developed a golf course for recreation and finance its maintenance. Membership fees are affordable enough to accommodate modest wage earners such as industrial laborers and chauffeurs. Volunteers have donated their labour and materials to develop this recreational outlet which, in turn, benefits Bogota with its many green acres and wetlands. (IDB 1990).

Education Parks and other green areas also provide educational opportunities for urban residents. Many cities in Latin America boast botanical gardens, zoos, nature trails and even visitor information centres that can inform residents and tourists alike about the areas flora and fauna. Individuals, families and school groups can take advantage of a city's green areas to learn about the environment and natural processes. For urban children, as well as adult students, the learning experiences available in urban parks may be some of the few opportunities they have to learn

about nature through first-hand experience. Moreover, by getting the public involved in educational activities associated with urban green spaces, planners can raise the consciousness of the public concerning the importance of these spaces. Another way to educate the public on the importance and benefits of urban greening is to involve people in the greening process itself. Examples of this can be found throughout Latin America. In Sao Paulo, Brazil, Projector Artificer made use of the city's environmental secretariat to distribute 2000 trees to every elementary school in the urban area. Each child then received at least one tree in a folder to take home to plant in the yard. The project's objective is for the children to educate and motivate their families about the need to participate in the city's environmental programs. Similarly, a

Aesthetics

While not considered as important as filling basic needs such as food and shelter, the aesthetics of green areas can also be very meaningful to many urban residents. There are many examples of civic groups sponsoring the planting of trees to improve the aesthetics of key parts of their cities in order to increase civic pride. Vegetation reduces sun glare and reflection, complements architectural features and tones down the harshness of large expanses of concrete. Areas of a city with enough greenery to be aesthetically pleasing, are attractive to residents and investors alike. The beautification of Singapore and Kuala Lumpur, Malaysia, was one of the factors that attracted significant foreign investment that assisted those cities' rapid economic growth (Braatz 1993). In the Black Country district of England, urban greening is focused along roadways and railway lines in an attempt to improve the area's image and attract investment.

Another advantage of aesthetically pleasing green areas is their positive effect on property values. When unsightly vacant lots or garbage dumps are replaced by attractive parks, not only does the residents' quality of life improve, but the value of their property increases. In addition, rehabilitating lands with vegetation is often more attractive and cost-effective than constructing new buildings on them. Urban greenery, used as a boundary line, can improve the aesthetics of small private land-holdings in urban or suburban areas. One of the most common sights in Latin America and the Caribbean is the home dwelling, however rich or poor, bordered by

hedges of greenery, flower pots or food plantings. On large areas of municipal lands, such borders may be maintained through community efforts by nearby residents who value the green areas and join their efforts in order to protect them.

1.11. Challenges Facing in urban green spaces

The benefits provided by green spaces according to the ecosystem services perspective are endangered by human actions, such as urbanization (Niemelä et al., 17 2012). As the development of new policies focuses on managing this increasing growth, such as policies aiming for the densification of cities in order to combat urban sprawl and its problems, (Fuller et al., 2009; Jansson, 2014; James et al., 2009; Sandström, 2002; Niemelä et al., 2012). In many European countries, the recent changes in land use and life style favour the development of single and semi-detached houses far away from centres of services and business, causing longer commuting distances, so that new urban development's use increasingly more land per habitant (Niemelä et al., 2012). Even if densification is sometimes seen as a 'sustainable' solution (Jansson, 2014), the presence of the green spaces is threatened, and the capacity these spaces have to produce ecosystem services is endangered (de la Barrera et al., 2015; Niemelä et al., 2012).

Literature doesn't argue against urban densification, but instead argues for a similar densification of the green structure, in concordance to the settlement's structure and configuration, so that the benefits of these areas still echo in the population (Jansson, 2014). In addition, the expansion of urban areas has resulted, in most cases, in the 'fragmentation and isolation of urban green spaces from rural green areas' (Niemelä et al., 2012, p 3226), causing the development of a heterogenic area that hinders these areas to provide ecosystem services to their maximum capacity (Andersson et al., 2014). This might also result in more pressure on existing green spaces, as they are demanded to have a multifunctionality that endangers the ecosystem services they provide with multiple objectives, conflicting interests and meanings (Andesson et al., 2014).

Various authors point towards these green spaces as mechanisms to fight the negative effects of this intense urbanization (Fuller et al., 2009; Jansson, 2014; Maheswaran, 2010), as it has been mentioned above. It is in this context that appears the necessity to study the way in which a city is managing these green spaces. What are considered green spaces? How do these green areas distribute within the city? And how well are they connected to

the citizens that should benefit from their presence? How are the local and national planning practices taking these issues into account? These questions seemed to need an answer to fully understand the role these green spaces play, and the role they will have in the future; and to understand how different cities have approached this issue.

Developing institutional capacity, using appropriate technology and securing sustainable funding from a variety of sources are also crucial yet elusive goals. Another important challenge is maximizing public participation at all levels of project implementation, and breaking down the barriers that prevent this, including lack of land tenure, inequality in participation between the sexes, and outdated legislation. Finally, planners face the hurdle of greening areas that have experienced moderate to severe environmental degradation. Overcoming this problem may require a multi-tiered approach involving cooperation with other government sectors to reduce deleterious impacts, land reclamation and strict monitoring procedures. This chapter describes these challenges in more detail using examples from case studies to illustrate the importance of overcoming such barriers to success.

Economic Valuation for Urban Greening Placing a value on the city's green resources is one of the most significant challenges that city planners may encounter when implementing an urban greening program. While there are obvious costs involved in the establishment and upkeep of green areas, it is difficult to calculate the value of all the benefits associated with that area. It may also be difficult to avoid double counting benefits if several assessment methods are used independently to estimate a variety of benefits from the green areas.

Green areas provide a range of tangible, easily valued benefits such as food, fuel and fodder from agricultural plots, but they also provide intangible yet valuable amenities such as aesthetics and noise reduction. How can city administrators value such multiple and diverse benefits and then compare them to benefits from other projects with more easily calculated costs and benefits? How does one place an economic value on something intangible? Can non-market benefits and ecological services be measured in the same units as market goods? If the government were willing to put land in green areas up for bid, with the restriction that it could only be used for urban greening purposes, the highest bid might be used as the economic value of green space. Equivalently, if the government acted as if it were a private developer, it could

achieve the same outcome by calculating the cash result of using the land in different green activities and choosing the most profitable. These procedures would be reasonable if the bids or the government's calculations reflected all costs and benefits. Unfortunately, this is not likely to be the case. With the exception of the goods produced by urban agriculture, many of the goods and services provided by urban greening activities produce no cash revenues.

They produce public goods (cleaner air, a prettier view, a quieter environment) that are enjoyed by many, but that cannot be charged to each user since people cannot be excluded from enjoying them. Economic analysis of urban greening initiatives is particularly relevant when sizeable tracts of urban land are involved. Such land has high value in private use because it can yield commercial and residential services that are precluded when the land is set aside for urban greening. The loss must be compared to the gain.

Furthermore, the government will incur large recurrent costs maintaining the greening activities. It is important to demonstrate that greening benefits (e.g., aesthetic improvements, wildlife habitat provision, etc.) more than offset the opportunity and cash flow costs of maintaining space as green areas. Thus, urban greening projects should be considered public investment projects and should be evaluated using economic analysis criteria

The recreational benefits of public parks that attract visitors from near and far can be approximated using a travel cost model. The number of trips taken over a season by a sample of visitors to an existing site (that is similar to the new, proposed site) will generally fall as travel cost rises. Assuming that visitors would react to a hypothetical fee in the same way they react to actual out-of-pocket travel expenses allows the analyst to derive a demand curve (price-quantity relationship) from the travel cost data. The total willingness to pay (gross benefit) is the area under the curve. This relationship breaks down for small urban parks where travel distances are short, or for new parks that do not have an existing close substitute for comparison.

A less satisfactory approach is to multiply the expected number of annual visits to the park by user day values per visit, based on expert judgement or averages of values reported in a large number of individual studies. Such values may not exist for much of Latin America. Air pollution reduction benefits have often been

obtained by applying a hedonic model that statistically captures the relationship between property values and the characteristics of the housing unit and the surrounding neighbourhood. Neighbourhood characteristics can include particulate and sulphur dioxide concentrations, traffic noise and other ambient quality measures. The estimated increase in property value resulting from decreases in one or more air pollutants brought about by an urban greening effort is a proxy for the expected benefits

1.12. Basic Requirements for Developing an Urban Greening Program

Types of Projects There are three principal investment designs for urban greening programs, depending on the number and size of the cities involved, and on the priority of greening among the investment projects objectives. The three types are (1) individual self-sustaining projects, (2) multiple works projects, and (3) sub component projects; they are described below. Individual projects are primarily for major cities and metropolitan areas that require enough funding to merit consideration as individual investment projects. Large amounts of infrastructure development, technical assistance and mobilization of resources make it economical to approach a citywide greening effort as a single project. Multiple works projects fund urban greening efforts for more than one city or region at a time. Such projects are usually for smaller urban centres that would be costly to fund on an individual basis, but when combined make efficient use of resources. Greening programs of this type may share resources, using similarly formatted approaches that work toward more general, and less city-specific solutions. As a component of integrated city projects, urban greening can be accomplished without going through the process of instituting an exclusive funding project. For example, this method might incorporate urban greening into water and sewer projects by requiring additional forested land around settling pools and reservoirs to serve as a catchment area, or requiring greenways along aqueducts to protect flood plains. While this strategy generally makes smaller contributions toward a city's urban greening than the above methods, it may be much easier and faster to approve and effectuate. As much as this integration method facilitates additions to a city's greenspaces, urban greening components should still only be included to the extent that their economic returns warrant and that they contribute to the overall success of the project.

- Environmental Education

- Institutional Strengthening
- Maps
- Social and Political Support Social Support
- Financial and Economic Viability
- Activities of Urban Greening Programs
- Public Outreach and Education
- Public Parks
- Watershed Management
- Financing Urban Greening
- Potential Funding Strategies
- Cost Avoidance, Reduction and Recovery
- Trust Funds

1.13. URBAN GREEN GROWTH STRATEGIES FOR INDIAN CITIES

India has to leverage its cities despite the fact that an overwhelming majority of them are deficient in basic services, financially weak and lacking the capacity required to plan and implement change. Growing at a decadal rate of 31.76% (2001-2011), these urban areas house almost 377 million people and are expected to accommodate 590 million people by 2030. Urban growth is far exceeding the capacity of infrastructure and services, and inadequate environmental management measures have contributed to a significant degradation of valuable natural resources, adversely affecting the quality of life of urban dwellers. Conventional urban planning and management tools have become barriers in their development and in order for this growth to be sustainable, there is a desperate need for new policies and implementation strategies that can generate positive social, environmental and economic impact.

- About 27.8% of population are residing in urban areas in India.
- There are 6 mega cities (4 million plus) & 35 metropolitan cities (1 million plus). There are also, 393 large towns (0.1 million plus) and 3,784 small and medium towns (less than 0.1 million population).

- In the context of carbon offsets, massive greening is required in urban areas to provide shade and insulation to buildings. Urban greening plays a vital role in the

Table 1: Table Showing Pattern of Urbanization in India

Census Year	Urban Population (in millions)	Percentage Urban	Annual Exponential urban growth rate
1961	78.94	17.97	-
1971	109.11	19.91	3.23
1981	159.46	23.34	3.79
1991	217.18	25.72	3.09
2001	286.12	27.86	2.75
2011	377.10	31.16	2.76

Source: Bhagat, 2011

Figure 6: Urbanization in India

amelioration of urban environment. population increment of 90.5 million for the first time since independence (Bhagat, 2011) and is expected to hit figure of 550 million or 42% of the total population by 2030 (Roberts and Kanaley, 2006).

- With growing cities the demand of resources like water, land etc. has grown proportionally to the growing rate of the urban population. Hence, Cities are experiencing increasing signs of environmental stress, notably in the form of poor air quality, excessive noise, and traffic congestion (Ruangrit & Sokhi, 2004).
- At the same time, irregular and unsustainable extension of cities has caused destruction of urban green areas and resulted from increasing demand for land

1.14. MAJOR INDIAN CITIES WITH PER CAPITA GREEN SPACE.

The major impact of urbanization on landscape surfaces is the replacement of vegetated areas with artificial surfaces that are mostly impervious. These impacts modify the surface energy balance through a change in absorption and reflection of solar radiation. Since 1970, India has been undergoing rapid urbanization. The urban population rose from 109 million in 1971 to 377 million in 2011, by an 11.7% increase (from 19.9% to 31.6%) over four decades. The number of cities in the country with the population exceeding one million has steadily increased from 23 in 1991, and 35 in 2001, to 53 in 2011.

For instance, New Delhi, a megacity in a humid sub-tropical setting, has complex patterns of urbanization in terms of its commercial, residential, mixed-use areas, and traffic intersections. For this, the major contributing factors are high population density, high

density of road network, and a very high amount of traffic flow. Unlike Delhi, the city of Pune is located in a hot, semi-arid region with village nuclei and industrial sectors. Pune is affected by a sudden recent increase in population by 10 times compared to the last century

- ❖ Table shows the population of populous cities of 2018 Census which was collected from the India Environment and Population Portal [34]. The table also provides information about the geographical area, forest and tree cover, and per capita green space in these cities.
- ❖ It can be noted from that many cities in India already fall short of green space available per capita, which is much below the WHO recommended norms of 9 sq m/capita.
- ❖ Global standards recommend 33% green cover for urban areas, and that the maximum green spaces be connected.
- ❖ The LEED ND (Leadership in Energy and Environment Design Neighborhood development) recommends that the green area per capita be >20 sq m or maintain a minimum of 1.25 ha of open space per 1000 residents, and that the access to open space be within 250 m of residential areas for smaller parks

Table 1. Major cities of India with per capita green space.

City	Population in Millions	Population Density (km ⁻²)	Geographical Area (km ²)	Green Cover % (in km ² ; 2017)	Per Capita Green Space (m ² ; 2018)
Delhi	28.50	12,591	1484.00	20.00 (296.80)	10.41
Mumbai	23.50	20,482	603.00	36.48 (220.00)	9.36
Kolkata	15.20	24,400	1380.00	7.30 (100.74)	6.61
Bangalore	13.90	4381	2196.00	2.09 (46.03)	3.31
Hyderabad	11.57	18,480	650.00	1.66 (10.79)	0.93
Chennai	9.88	14,350	1189.00	15.00 (178.35)	18.05
Ahmedabad	8.41	9900	464.00	17.00 (78.88)	9.38
Surat	6.55	1376	326.50	11.84 (38.66)	5.90
Gandhinagar	6.33	660	649.00	54.00(188.46)	29.77
Jaipur	3.71	598	467.00	5.43 (24.75)	6.67
Nagpur	2.94	11,000	285.90	18.00 (51.42)	17.49
Mysore	1.70	6911	128.40	20.19 (25.92)	15.25
Chandigarh	1.05	9252	114.00	35.00(39.90)	38.00

Note: Figures in bold indicate the above optimal of 9 m² per capita green cover.

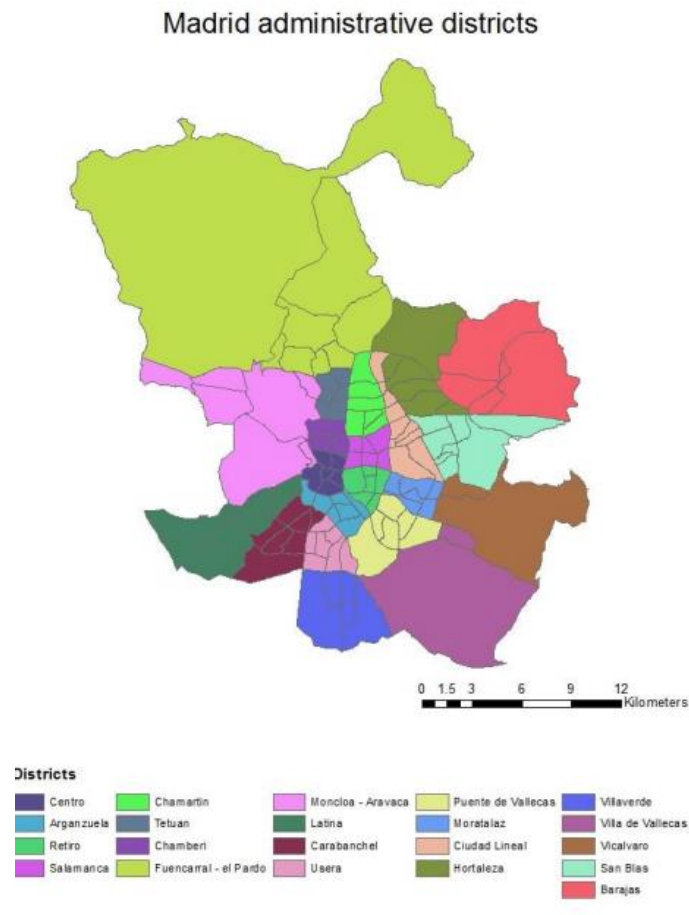
Figure 7: Per capita green spaces

2. DESKTOP STUD

2.1. International desktop study

Case study: Urban green cities of Europe

Madrid:



Map showing the distribution of the districts within the limits of the city of Madrid. As it can be seen, the city is divided in 21 different districts. Data source: Portal de Datos del Ayuntamiento de Madrid

Figure 8: Madrid city Map

2.2. History of Green Spaces in Madrid:

Madrid is a city located in the mountainous region of the center of Spain. The city is characterized by the natural areas around its limits, parts of bigger green structures. This green natural areas are part of protected natural parks: On the northwest of Madrid, the area of El Pardo and Soto Viñuales; and on the southeast, the Parque Regional del. The presence of these green natural areas of such a large ecological value within the limits of the city is a uniqueness of Madrid hard to find in another cities of Europe.

The rest of green spaces within the city are situated in the urban area, located in outer areas of the city, leaving the inner city 'empty' from these spaces. Madrid presents an interesting mixture of green spaces created recently, following modern trends and designed with a clear social purpose, and parks and green areas that were created only for royal recreation. The biggest green urban areas in the city belong to this last group of spaces. These areas, such as the Retiro park, or Los Jardines del Capricho, are catalogued as 'historic parks' in the legislation and were opened to the general public at the ending of the 19 century.

2.2.1. Green Spaces in planning practices in Madrid:

The planning system in the city of Madrid is defined by the General Plan of Urban planning of Madrid. This plan was developed in 1997, and has been applied since then, with the addition of some modifications to the original regulations. Nevertheless, the main ideas present in this plan have been mostly unchanged until nowadays. A new document in regards to the planning of green spaces was designed and completed in 2013, the Plan Revision for Urban Sustainability. Including of the green regions , developing the green spaces the quality of the urban landscape

In addition, it is important to mention that green spaces within the city (also sometimes called 'parks') are classified in different groups according to their values, year of development, etc. These categories are: Urban parks (Parques rbanos), Historical Parks (Parques históricos) and Forestal parks (Parques forestales), each with specific conservation and planning regulations.

Background information about Madrid

In this section, information about the administrative boundaries of Madrid, as well as about the distribution of the density of population in the city and the presence of both green urban

Urban Green Areas in Madrid

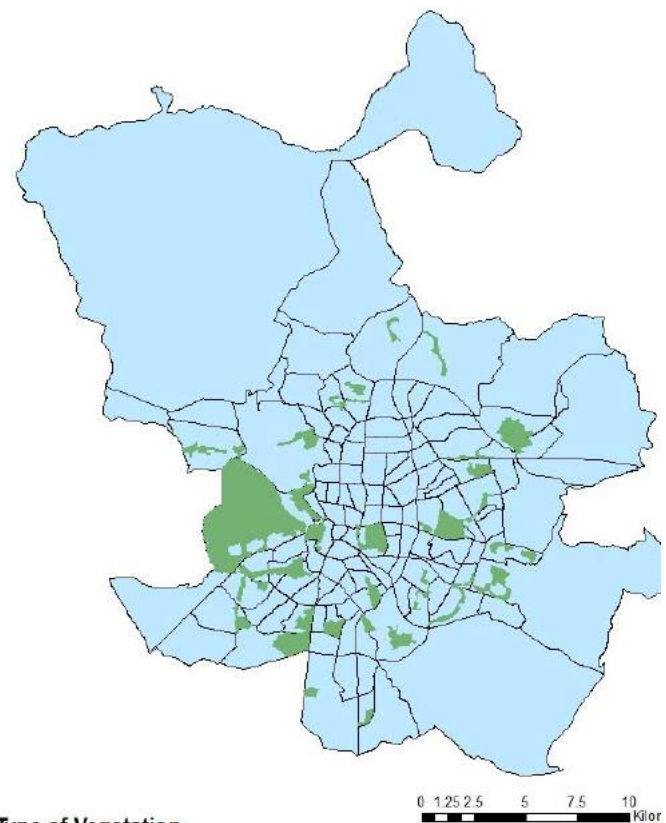


Figure 9: Urban green areas in Madrid

spaces and green natural spaces is given. This information will be used later on this paper to illustrate and reflect upon the results obtained.

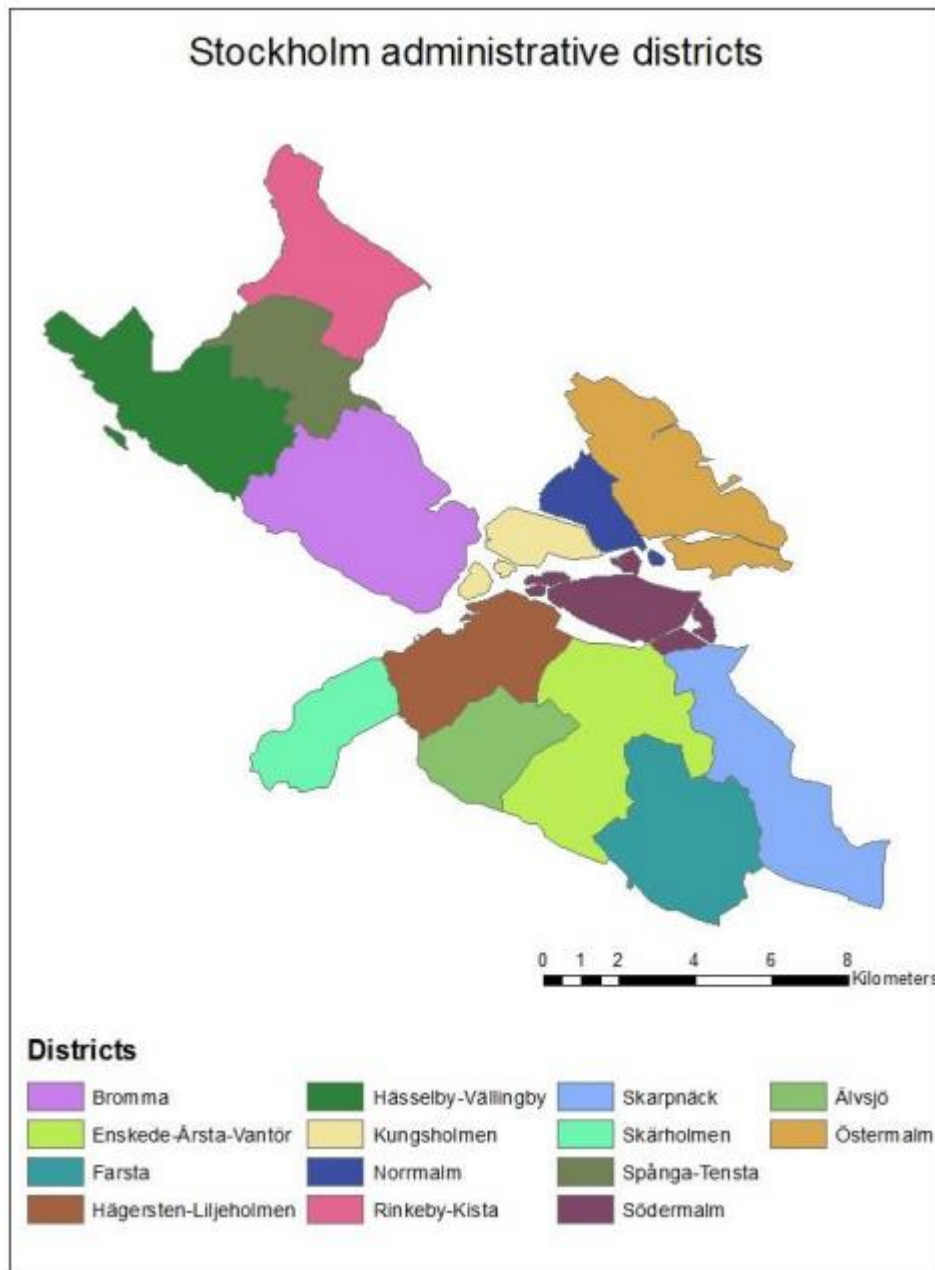
Distribution of green urban areas within the city limits. Data source: Corine Land Cover (accessed March, 2017)

The presence of green spaces in the city is also the results of previous actions taken by the government. This is for example the case of the green wedges ('cuñas verdes' in Spanish) that were designed and planned in the 1980s. These wedges were not all established, and some of them were later removed to leave space for urban growth (Area de Gobierno de Urbanismo y Vivienda, 2013). Even though the lack of presence of green spaces in the inner parts of the city is clear, the documents highlight the presence of trees in the whole city, as to 'compensate' this lack of green spaces (Ayuntamiento de Madrid, 2009; Area de Gobierno de Urbanismo y Vivienda, 2013). Other elements that can be addressed as green spaces are the gardenized roundabouts that are present in some traffic axis in the city. However, both the presence of trees and these roundabouts do not guarantee any influence on the population, since they are traffic areas, and they are only experience for a short period of time

2.3. Green Spaces in Stockholm:

Background information about Stockholm :

As it has been done in the previous section about Madrid, information about the administrative boundaries, the distribution of the density of population in the city and the distribution of green spaces (both urban and 'natural') will be presented in this section.



Map showing the distribution of the districts of Stockholm.
The city is divided in 14 district councils

Figure 10: Stockholm District Map

2.4. History of Green spaces in Stockholm

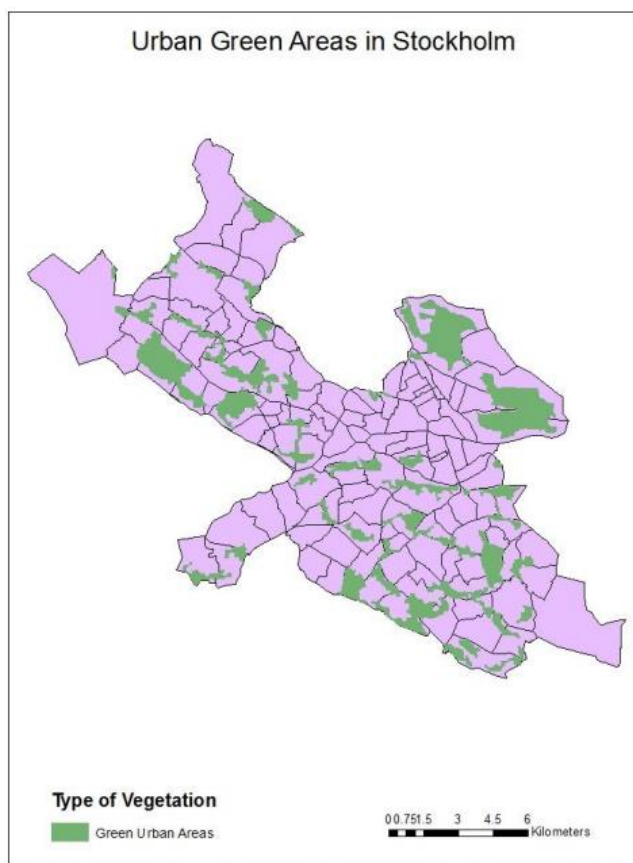


Figure 11: Urban Green spaces in stockholm

Distribution of green urban areas within the city of Stockholm. Data source: Corine Land Cover (accessed March, 2017)

Stockholm presents a unique city environment. The city is characterized by the presence of three large bodies of water (Saltsjön, the inner city and lake Mälaren), and for its growth in defined circles around the old town area, although in some parts of the city, especially in the outer ones, these circles are interrupted by large infrastructures and elements such as green spaces (Stockholm City Council, 2010). These growth rings represent the strong planning tradition this city has had, leaving traces of different 33 planning currents from the Age of Greatness and the industrialisation, to garden cities or 1950s suburbs (Stockholm City Council, 2010).

This strong presence of planning culture in the grid of the city translates in the expansion of urban districts following a slightly rigid regulates pattern (Stockholm City Council, 2010). The city of Stockholm has several green spaces that were created following this trends of growth, along with planning directives and currents of thought. Various of these examples can still be found in the city.

Around 1700 the main inspiration for green spaces were the English landscape parks, characterized by natural- romantic forms, an example of this kind of park is Bellevue, located in Norrmalm district. From 1850 to 1880 public city parks were developed, with paths, trees and exotic plants (such as the park of Berzelii, located also in the Norrmalm district).

Around 1900, the tendency was to develop parks following romantic currents, with rocks and domestic trees and plants (Kronobergsparken, in

Kungsholmen). Around 1920s the geometric figures from the classicalism were introduced in the parks, inspired by the tight house architecture (Ellen Keys park, in Skärholmen district). Functionalism became the main trend from 1930- 1950, with simple and landscaped, with playgrounds and another amenities.

Around 1960-1970, late modernism was translated into the importance of large scale, with big grasslands and single minded plant materials. From 1980-1990 Stockholm was densified, and parks were established, in form of postmodern poetic and formalist parks.

Green Spaces in planning practices in Stockholm As it has been mentioned previously, Stockholm's City policies have been working towards a sustainable development, branding the city as 'green' within the international context. This aspect also affects that way in which green spaces are treated in this city. Following the guidelines set in Rio Earth Summit (1992), Swedish legislations were remodelled towards a more sustainable development, including the management of green spaces in and surrounding urban areas (Sandström, 2002).

Actual legislation establishes that Swedish towns and cities are encouraged to develop green plans as an important part of the compulsory structure plan (Sandström, 2002). The focus of this green plan is to identify urban green spaces and highlight their value to the community, in order to try to achieve sustainability and to preserve biodiversity (Sandström, 2002) 34 This main focus is present in Stockholm urban plans, both past and present.

For example, as it is established in 'The Walkable City' City Plan, "Development of the inner city and the strategic nodes in the outer city, linking these with new public transport and establishing more cohesive urban environments and more attractive parks and green spaces will lead to a city that is sustainable in every way" (Stockholm City Council, 2010, p. 9) It can therefore be seen the principal role these green spaces are given in the configuration of the city, and in its international branding policy.

Even though this Plan is not legally binding, it sets the guidelines for other, more detailed plans, as it was mentioned previously. The importance of this plan relies on the fact that, because Sweden has what is known as a 'municipal planning monopoly', which means that it is the City of Stockholm (being a municipality) that is in charge of drawing up and adopting detailed development plans within its boundaries.

The previous city plan proved the need to value the benefits green spaces provide, as well as the need for a more holistic perspective in order to guarantee that any future development has a positive impact on the urban environment (Stockholm City Council, 2010), as the documents states, ' the city's green spaces are to be developed so that Stockholmers can enjoy them fully, and new parks are to be created and established' (Stockholm City Council, 2010, p 3).

It also stressed the need to maintain all the elements present in the contiguous green structure, an approach that has changed in recent years on behalf of a more realistic solution, that matches the need for expansion of the city while maintaining certain green standards (Stockholm City Council, 2010). This plan establishes that, given the amount of green spaces and the closeness to nature the city has, 'access to parks, sports facilities, cemeteries and countryside is a strong contributor to Stockholm's attractiveness, character and beauty' (Stockholm City Council, 2010), p 20).

This present Plan also reflects upon some of the ecosystem services green spaces can provide to a city. For example, it refers to the relationship between the existence of these spaces and the improvement of social cohesion (reflecting also upon matters such as the need to consider different population groups as users of these areas, an issue of high importance for maximizing the possible benefits these areas have, as has been discussed previously), or the connection between the presence of accessible green spaces and physical activity

'Today streets, parks and squares of the inner city function partly as shared spaces for many Stockholmers, wherever they may live'

The 'Parkprogram' focuses on the green spaces located within the city, as well as accessibility to parks and the countryside (Stockholm City Council, 2010). It is important to highlight the fact that this program focuses on the richness of these areas, assuming the fact that the number of this areas will decrease in the future. The main purpose of the Parkprogram is to create a consensus in what is essential for a city's park and natural area, and how it should be handled (Markkontoret, 2006). Through this document, green spaces within the city are generally referred to as 'parks', although there is a description of the consideration of each type of green space according to its shape, dimensions, etc. In this program, measures regarding the planning, the different projects (created when there is a

need or desire to change parks or natural areas) and the maintenance of these areas are established (Markkontoret, 2006).

There are several plans and international conference that have had an impact in the planning of these areas, such as Agenda 21, Barnkonventionen, Stockholms Grönkarta or EU directives (Markkontoret, 2006). All these plans were included in the development of the Parkprogram. This document mentions some of the ecosystem services these areas provide, as values of the presence of these areas in the city, valuing the effects different kind of green spaces have, including elements such as football fields or bathing spaces, and classifying them from areas with values related to health, to areas with cultural or historic values (some parks, like Kungsträdgården, are several centuries old, and therefore their importance also relies on their historical value (Markkontoret, 2006)), with interesting landscape forms, identity or biological values, etc. (Markkontoret, 2006). Even if the ecosystem approach is not directly mentioned, several of the aspects this approach includes, such as benefits in relation with health, biodiversity, social interaction, etc., are included in the document.

RESULTS

In this section, results obtained following the methods mentioned above are presented. The results will be shown in form of maps, and table

2.4.1. Results obtained from Madrid

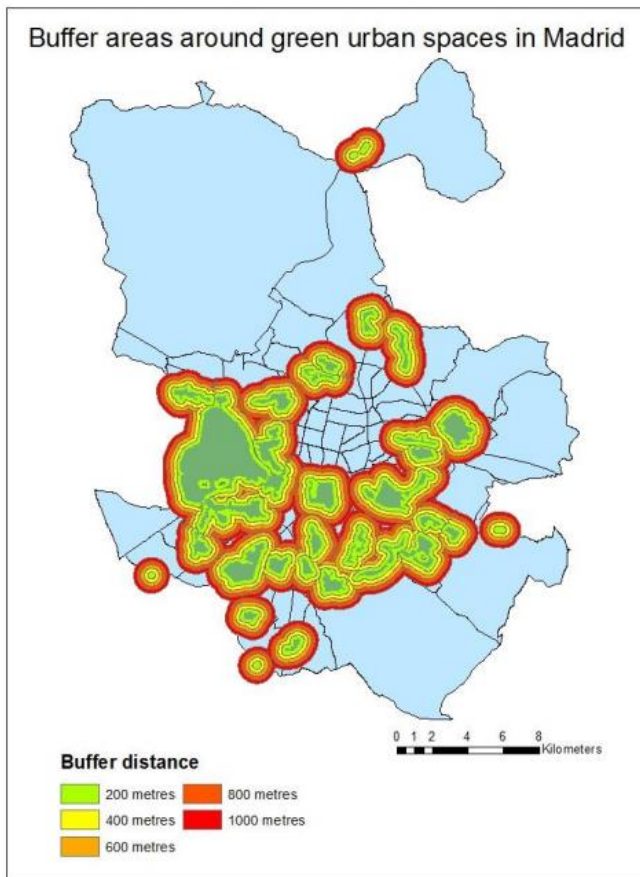


Figure showing the distribution of green spaces (both 'urban' and 'natural') in relation with the whole extension of the area of Madrid, in order to measure the 'quantity' of these areas in the city.

Map showing the influence of green urban areas in the form of buffer areas, this is, the accessibility of these areas.

Figure 12: Buffer areas around urban spaces

	Total Area (ha)	% Occupied by Green Spaces
Green Urban Areas (ha)	3573.389195	5.911767141
Green Natural Areas (ha)	28842.24064	47.71621594
Total Area Madrid (ha)	60445.36447	53.62798308

Table 2: Total area and % of Green spaces

Buffer Distance	Population	% of Population
200 metres	647562,7	20%
400 metres	1078650	33%
600 metres	1495676	47%
800 metres	1872334	58%
1000 metres	2166897	68%

Table 1: Buffer distance

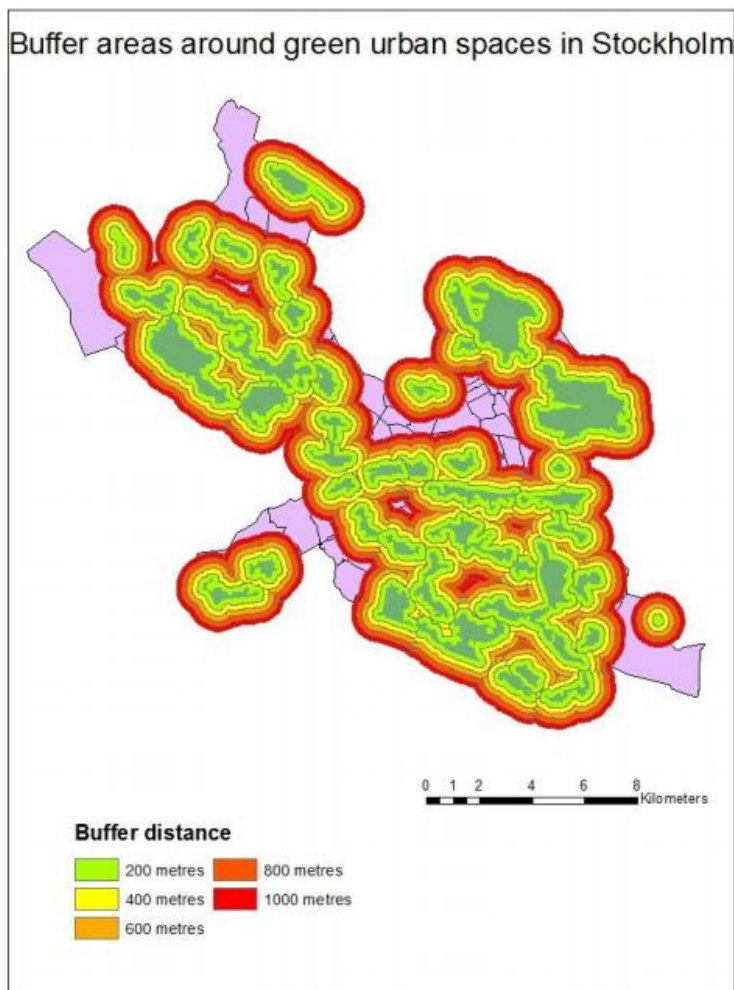
Figure showing the amount of population located in each buffer area from urban green spaces in Madrid.

2.4.2. Results obtained from Stockholm

	Total Area (ha)	% Occupied by Green Spaces
Green Urban Areas (ha)	3723.457448	17.25333378
Green Natural Areas (ha)	2013.74	9.33103946
Total Area Stockholm (ha)	21581.08975	26.58437324

Table 3: Total area and % of green spaces Stockholm

Figure showing the distribution of green spaces (both 'urban' and 'natural') in relation with the whole extension of the area of Stockholm



Map showing the influence of green urban areas in the form of buffer areas in Stockholm, this is, their accessibility.

Figure 13: Buffer areas around urban spaces in stockholm

2.4.3. Analysis of results of Madrid

In Madrid, the presence of urban green spaces was low, and they are located as a ring around the city, there are several areas in the city that lack the presence of this kind of areas, such as the centre of the city (which also happens to be the area in which the population density is higher), and the outskirts of the city limits (see Maps 4, 6 and 8). When

it comes to green natural spaces, Madrid has one district (Fuencarral- el Pardo) which is mainly a natural space that has been preserved mostly unchanged because it belongs to the King of Spain. This characteristic may contribute to alter the results obtained in this study, as it will be discussed later on. Apart from this district, Madrid is connected in its southern part with a ecological natural park, Parque Regional del Sureste. These two areas offer contact with natural landscapes to the habitants in the city.

2.4.4. Analysis of results of Stockholm

Stockholm has a green structure that varies widely from the one present in Madrid. The city counts with several green urban spaces scattered across the map, being a total of 17.25% of the whole territory (Fig 5). On the other hand, the natural green spaces are less common, and are mostly limited to the north and south borders of the city, being a 9.33% of the territory (Fig. 5), so that 26.58% of the whole territory is covered in some kind of green space.

While studying Stockholm it is necessary to reflect upon the role waterways and water bodies play in this area. Even though they were not considered 'green spaces', it is necessary to comment on them, since they can provide with some ecosystem services to the city, such as stress relief improving social cohesion. This represents a duality: on one hand, it could serve as to increase some of the benefits of green spaces in the city. On the other hand, the presence of these water bodies could in fact reduce accessibility to green spaces

Good practices in madrid and stockholm Spain:

The Barcelona Green Infrastructure and Biodiversity Plan 2020 Barcelona is one of the most densely populated cities in Europe. Whilst it has a relatively low ratio of green space per inhabitant, it has more street trees than most other European cities: currently around 161,423 trees of 150 different species line the streets of Barcelona. The 'Barcelona Green Infrastructure and Biodiversity Plan 2020' which was released in 2013, lists over 70 projects and actions provide environmental and social services, introduce nature into the city, increase biodiversity, increase connectivity among patchy Green Infrastructure and make the city more resilient.

Benefits of the project include:

- Trees have a positive impact on health since air pollution is a major environmental challenge for the city;

- Health and educational benefits from gardening and contact with nature

Ongoing Green Infrastructure projects and biodiversity projects are: Green Corridors Network, urban allotments with eco-gardening, habitat conservation (artificial reefs) and species conservation (protection of amphibians in urban ponds and protection of urban birds). There are artificial reefs in the Mediterranean Sea south of Barcelona to restore marine biodiversity; this pioneer project aims at facilitating the reproduction of species and benefitting traditional fishing methods.

2.2. A CASE STUDY OF PUNE, INDIA:

The main objective of the study to understand the social, ecological and economic values of urban green spaces. and deals with changes with urban Land Use / Land Cover Changes using remote sensing technique. For this purpose Landsat ETM+ images of 1999 and 2009 was obtained for the study area.

STUDY AREA :

Pune City (Pune municipal corporation limit), lies between latitudes 18°25' N and 18°37' N and 73°44' E

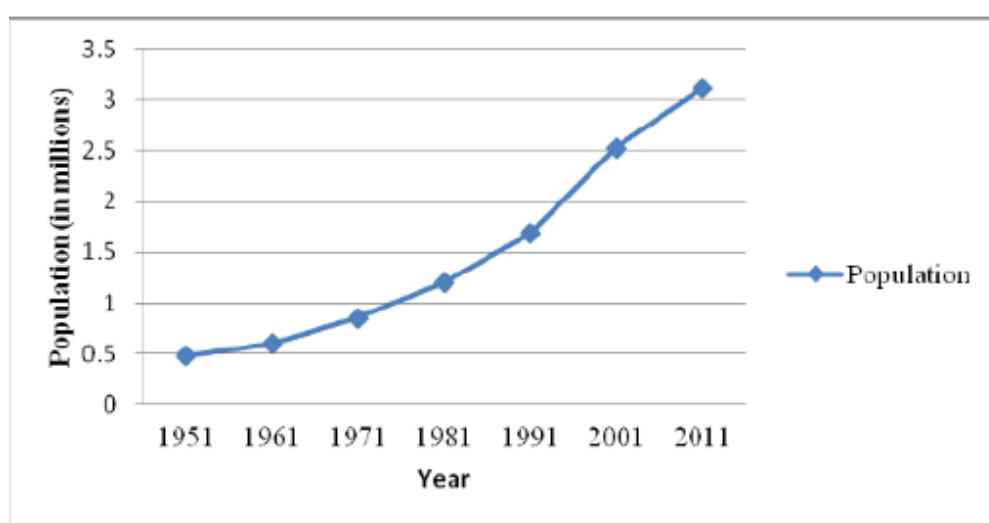
and 73° 57' E longitudes covering an area of 243.84 sq km. Situated on the Deccan plateau and lies on

leeward side of the Western Ghats and is at a height of 560m above the mean sea level

Table 3: Land Use/ Land Cover Changes in Pune City

Landuse Class	Area in 1999 (%)	Area in 2009 (%)	Area in 1999 (Sq. Km.)	Area in 2009 (Sq. Km.)	Area Difference (Sq. Km.)	Area Difference (%)
Water Bodies	1.47	1.20	3.58	2.93	-0.66	-0.27
Built Area	30.86	48.50	75.25	118.26	43.01	17.64
Barren & Fallow Land	36.20	21.80	88.27	53.16	-35.11	-14.40
Dense Vegetation	13.50	12.82	32.92	31.26	-1.66	-0.68
Sparse vegetation	17.97	15.68	43.82	38.23	-5.58	-2.29
Total Vegetation	31.47	28.50	76.74	69.49	-7.24	-2.97

Table 4: LanduseLandcover changes in pune city



Graphs 1: Population Data

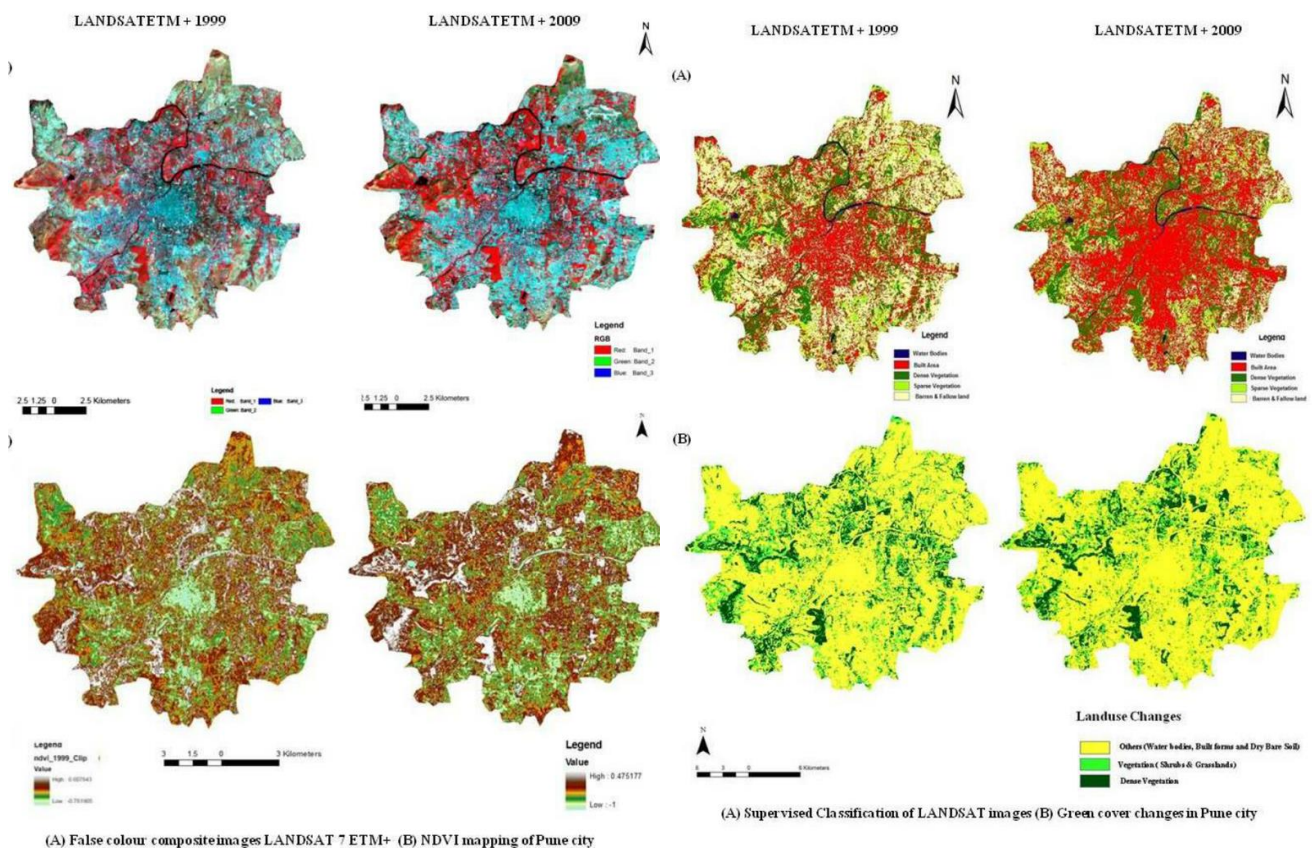
Pune's built up area has increased from 30.86% in 1999 to 48.50% in 2009 i.e. substantial increase of 43.01 sq km. of area in just last 10 years, on the other hand barren & fallow land area has decreased considerably from 36.20% in 1999 to 21.80 % in 2009.

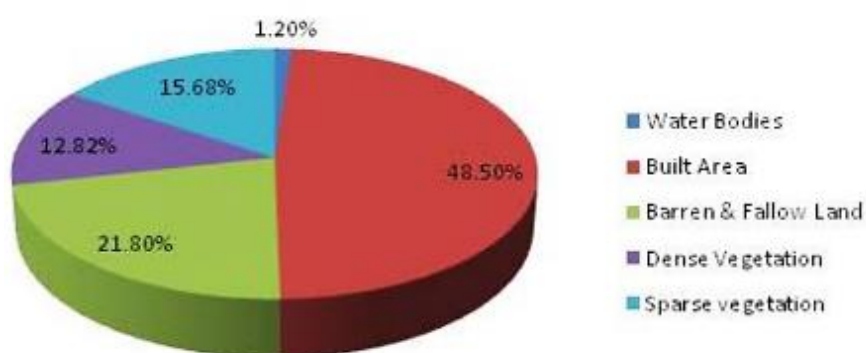
Major objective of the study was to analyse the changing trend of the urban green spaces in

the city, it was noticed that both sparse and dense vegetation showed decrease in areas of 5.58 and 1.66 sq km. respectively in past 10 years. Thus, this analysis highlights that the city is under high pressure of urbanization and sprawl, taking place by grabbing and converting existing habitat like agricultural, barren, fallow land & forestlands into built habitat and grey infrastructure. The overall mapping accuracy of the land use/ land cover map was 86.73%

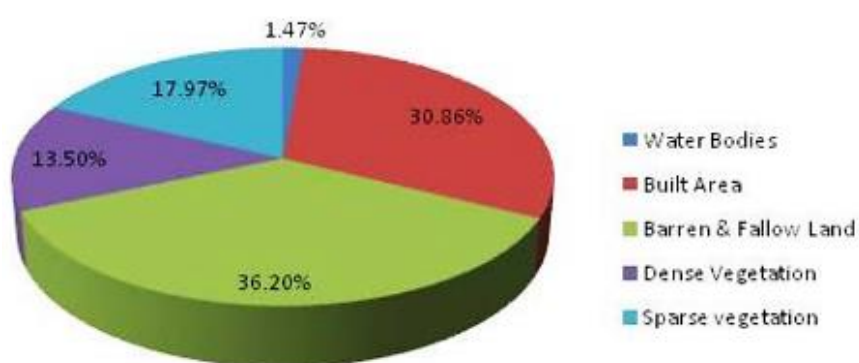
2.2.1. Landuse/Landcover Changes in Pune City

Presently city is experiencing large number of immigration from other regions of state and nation in search of economic opportunities, which is leading up to leapfrog increase in the city population





Graphs 2: Land use/landcover of 1999



Graphs 3: land use and landcover of Pune 2009

A) Land use/ landcover of Pune 1999 B) land use and landcover of Pune 2009

Graphical Representation
Land use/Landcover
Changes in Pune City

The three datasets with their parameters will be collated to prepare Comprehensive Green Space Map and Green Space Index for each ULB.

- Total of 141 ULBs in Telangana are evaluated.
- Some secondary data such as innovations, best practices, selection of tree species, landscape design shall be collected from ULBs to help identifying extra efforts by the ULBs in creating pleasing green environments.

National institute of urban management shall carry out the evaluation for green space index and furnish a report and ULBs will be awarded based on six different arenas of performances.

3. STUDY AREA

3.1. PROFILE:

SAIDABAD is a mandal falls under the GHMC(GREATER HYDERABAD MUNICIPAL COPORATION) boundary of Hyderabad the capital city of Telangana state. The city is one of the largest metropolitan of India with an area of 625.9 square kilometers (250 sq mi) with the population of 6,809,970 and further 7,749,334 living in the metropolitan area, making Hyderabad the fourth most populous city and the sixth-most populous urban agglomeration in India.

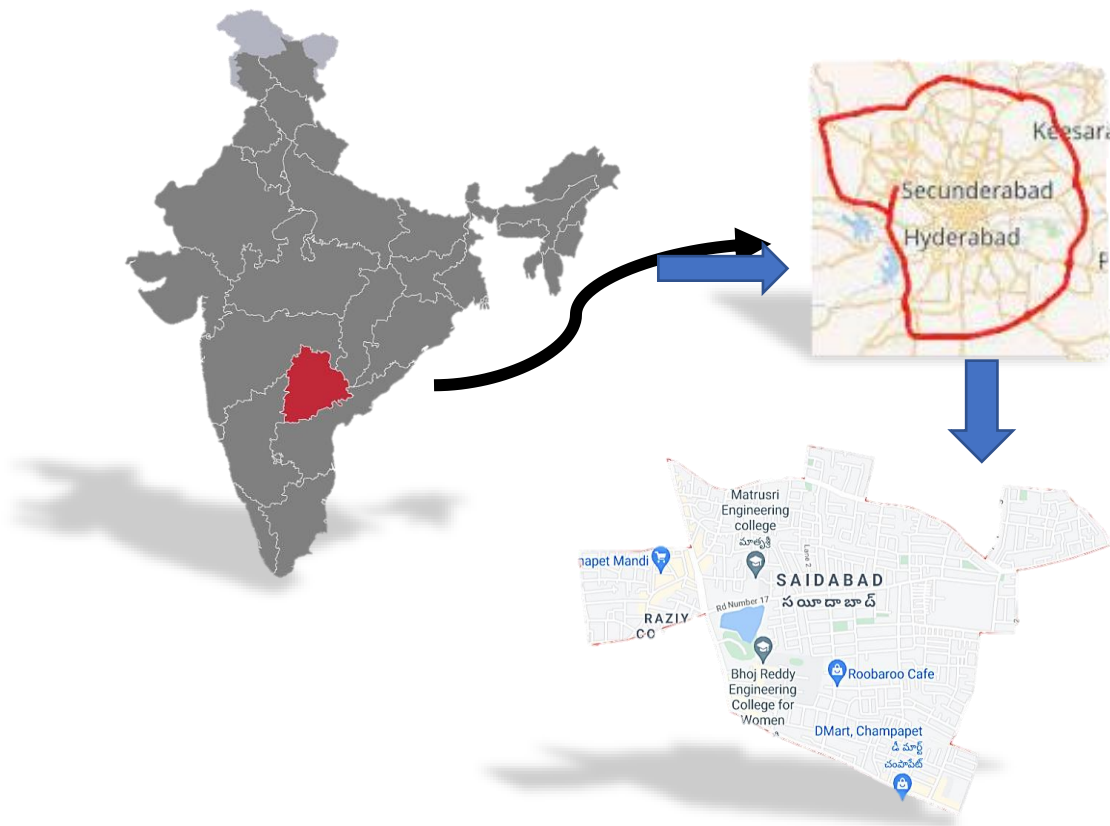


Figure 14: Location map of study area

3.2. DEMOGRAPHIC PROFILE OF STUDY AREA:

- AREA : 6.59sq KM
- total population : 345,722 as per the Census 2011.
- Male population : 177,222
- Female population : 168,500 In 2011 there were
- Total 74,462 families residing in Saidabad Mandal.

- Average Sex Ratio of Saidabad Mandal is 951.
- The average literacy rate in rural area is 84.7%
- Sex ratio : 951/1000 male
- The total literacy rate of Saidabad Mandal is 84.66%
- male literacy rate is 77.86%
- female literacy rate is 72.05%
- Population Density: 18098 people per km²

GHMC Ward No – 18, 19, 20, 21 Saidabad Mandal wards

As per the Population Census 2011 data, following are some quick facts about Saidabad Mandal.

	Total	Male	Female
Children (Age 0-6)	39,324	20,680	18,644
Literacy	84.66%	77.86%	72.05%
Scheduled Caste	20,336	10,400	9,936
Scheduled Tribe	5,809	3,068	2,741
Illiterate	86,335	39,244	47,091

Table 5: Demographic Data of study area

While the powers-that-be continue to twiddle their thumbs even in December, the city continues to grapple with the lack of quality green spaces. As per the figures of the Census 2011, the city has a population of 68.1 lakh and spans over 650 square kilometres. However, the figures, according to the GHMC website, show that the city has only 694.4 acres allocated for parks and open spaces, which is just 2.81 square kilometres. This means, on an average, there is only 2.7 square metres of open space per person in the city. As it turns out, the urban and regional development plans formulation and implementation (URDPFI) guidelines say that it is optimal to have at least 10-12 square metres per person. Even the national building code (NBC) prescribes that a person should be given at least three-square metres of organised green space.

“It is common knowledge that there is a lack of green spaces in the city. The parks that the GHMC looks to build, or has built, is not proportional to the population of the

city. We only have colony parks, which are used for cultural reasons. They are not spaces where anyone can walk in or exercise,” Prof. B. Sri Nagesh, department of geography, Osmania University, told Deccan Chronicle.

3.3. Timeline of the study area SAIDABAD 2000 – 2021 (10 years)



Figure 15:2000



Figure 16: 2010

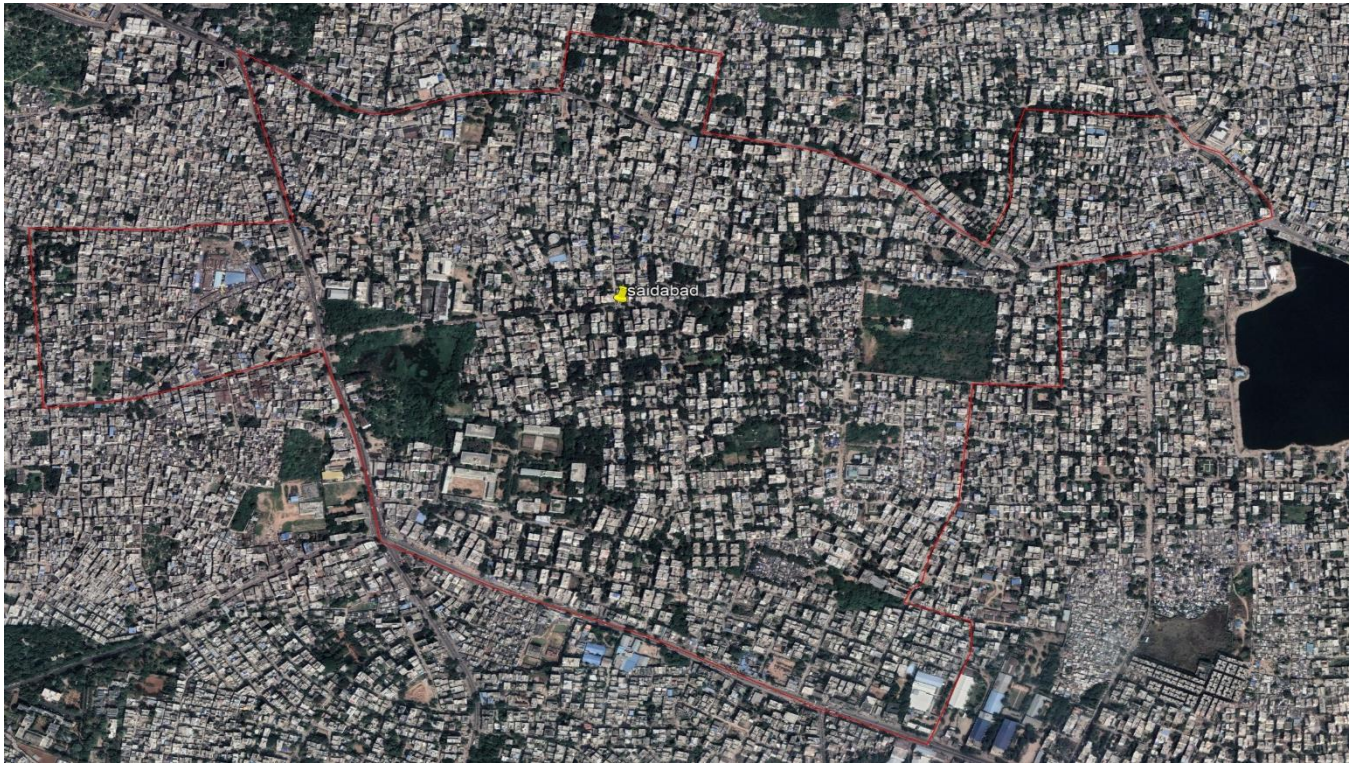


Figure 17:2020



Figure 18: Present state of the study area SAIDABAD ward of year 2021

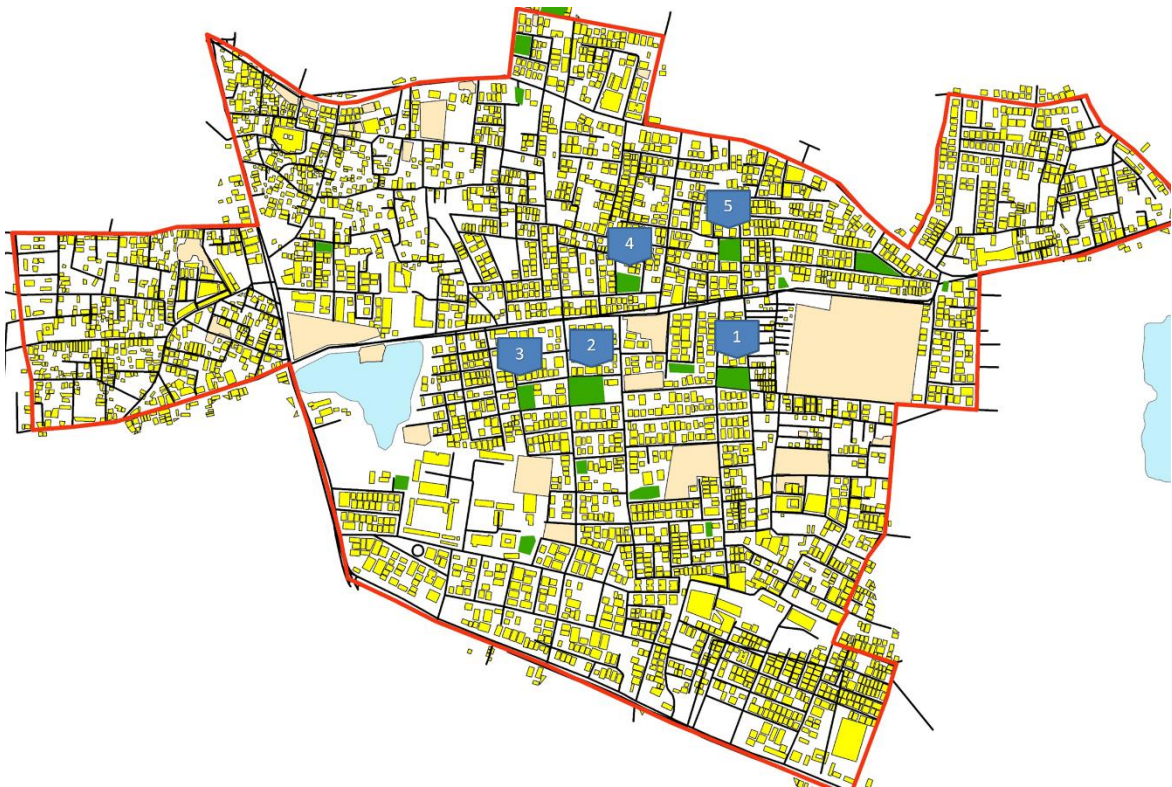
INFERENCE: The total study area consists of few green spaces – parks, open spaces with total area of 5.46 acres out of 603.11 acres of study area.

The green spaces present in the study area are community park, tree park.

GS COMMUNITY PARKS in study area:

1. Subramanya Nagar park
2. Lakshmi Nagar municipal park
3. Municipal park
4. Mahatma Gandhi park
5. Sri Srinivasa community park

The other small parts of land covered with shrubs wasteland.





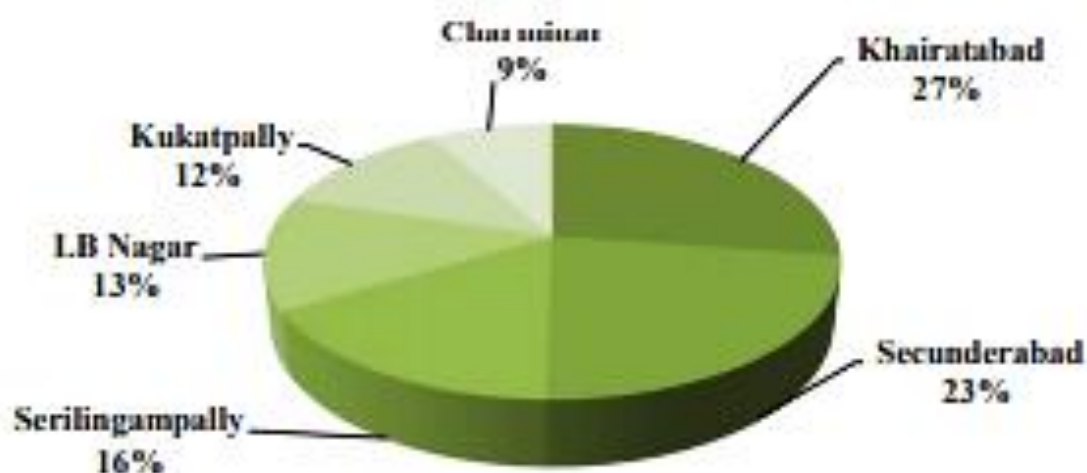
3.4. STUDY AREA JUSTIFICATION: PERCENTAGE OF PARKS WITHIN ZONES OF GHMC

S.no	ZONE	Population 2021 (in million)	Total ULB area (Sq. Kms)	Green cover (in Sq. Kms)	Green space provision per person (in sq.mts)
1	Charminar	1838679	98.694	28.65	0.12
2	Kukatpally	1206037	119.17	22.03	0.26
3	Khairatabad	1473880	78.063	17.25	0.49
4	Serilingampally	785196	136.258	21.09	0.55
5	LB Nagar	1168980	128.236	28.85	0.3
6	Secunderabad	1427228	65.41	16.01	0.43
	TOTAL	7.9 million	625.831	133.87	

Table 6: PERCENTAGE OF PARKS WITHIN ZONES OF GHMC

PARAMETER	PERCENTAGE	VALUE	QUALIFY CLASSES
Percentage of green in each cell	0-25	0.25	Low green
	25-50	0.50	Moderate green
	50-75	0.75	High green
	75-100	1	Very high

Table 7: Percentage of green



Graphs 4: Zone wise percentage of green

INFERENCE: GHMC has urban biodiversity resources such as major parks, colony parks, tree parks and traffic islands with a total area of 694.4 acres. There are about 159 traffic islands according to the GHMC. Khairatabad with 27% has the most number of parks, next followed by Secunderabad. Charminar has the least percentage in coverage of parks in all the zones.

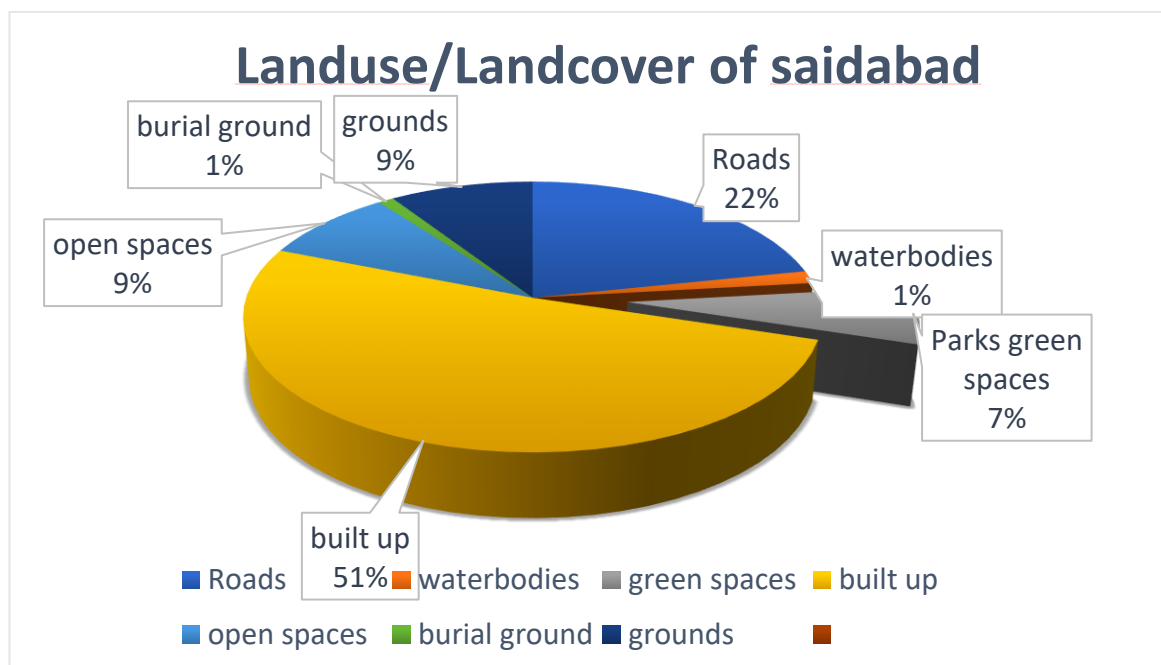
And Saidabad Mandal comes under Charminar circle which has 9% of parks.

Percentage of green in each cell is computed based on green area classification. The high resolution satellite image has been generated for the study area GHMC.

3.5. ANALYSING CLASSIFICATIONS OF PUBLIC URBAN GREEN SPACE

Public urban green spaces represent a large group of spatial concepts, forms, and functions. They are parks, streets, gardens, commons, in-between spaces, sports fields, cemeteries, etc. It can be discerned from the names of 69 forms that they are related to use, function, and form.

Based on the percentage, each cell has been classified into five green classes: very low, low, moderate, high and very high green quality on a scale of 0.2 – 1 where less than 20% of green in a cell has been categorized as very low quality green hence, given a value of 0.2. In the same manner, 0.4 (low green quality), 0.6 (moderate green quality) ,0.8 (high green quality) and 1 (very high green quality) value have been given to cells where percentage of green is 20-40%, 40-60% ,60- 80% and more than 80%, respectively



Graphs 5: Landuse/Landcover of saidabad

3.6. GREEN SPACE INDEX FOR GHMC

GREEN SPACE INDEX is a measure of green spaces available in particular neighborhood ward or city. The index will measure different parameters of green spaces .From the desk studies parameters identifies to develop green space index for GHMC are –

The green space index is shall be evaluated by collecting three different datasets

1. Green area map prepared using satellite imagery by GIS.
2. Green space provision per person through projection of population data through census.
3. Review of existing ongoing Urban greening projects through secondary survey by Urban biodiversity wing of GHMC

Land Use Category	Percentage of developed area			
	Small	Medium	Large	Metro Cities
Residential	45 - 50	40 - 45	35 - 40	35 - 40
Commercial	2 - 3	3 - 4	4 - 5	4 - 5
Industrial	8 - 10	8 - 10	10 - 12	12 - 14
Public & Semi-Public	6 - 8	10 - 12	12 - 14	14 - 16
Recreational	12 - 14	18 - 20	18 - 20	20 - 25
Transport & Communication	10 - 12	12 - 14	12 - 14	15 - 18
Agriculture & Water bodies	Balance	Balance	Balance	Balance

Source: UDPFI Guidelines

Table 8: Land use percentage for different developed areas

According to URDPFI guidelines, the landuse categories and percentage of development area is, for residential it is 45-50, 40-45, 35-40, 35-40 respective for small, medium , large and metro cities. for commercial it is 2-3, 3-4, 4-5 and 4-5 respectively. for industrial it is 8-10, 8-10,10-12,12-24 respectively. For recreation, it is 12-14, 18-20, 18-20, 20-25 respectively. for transportation and communication it is 10-12 , 12-14, 12-14, 15-18. the rest is allotted for agriculture and water bodies for the respective areas.

S no.	Parameters	percentage
1	ROADS	21.7
2	Water bodies	1.5
3	Parks (GS)	6.71
4	Built up	51.2
5	Open spaces	8.7
6	Burial ground	1
7	grounds	9.2
Total percentage		100

Table 9 - Study area landuse percentage

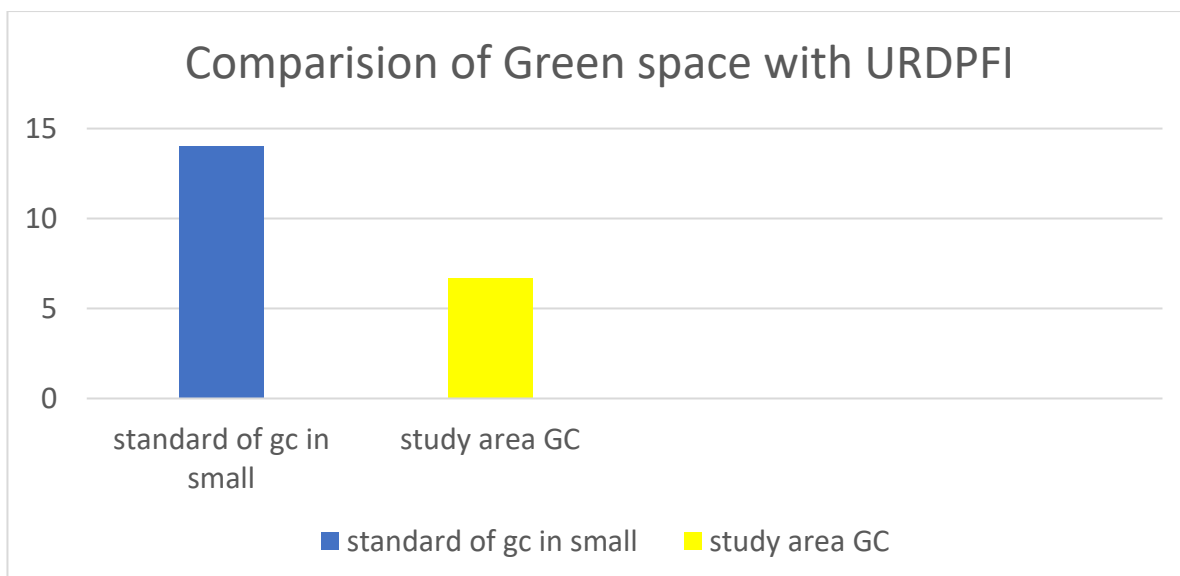


Table 10 Comparision of green space with URDPFI

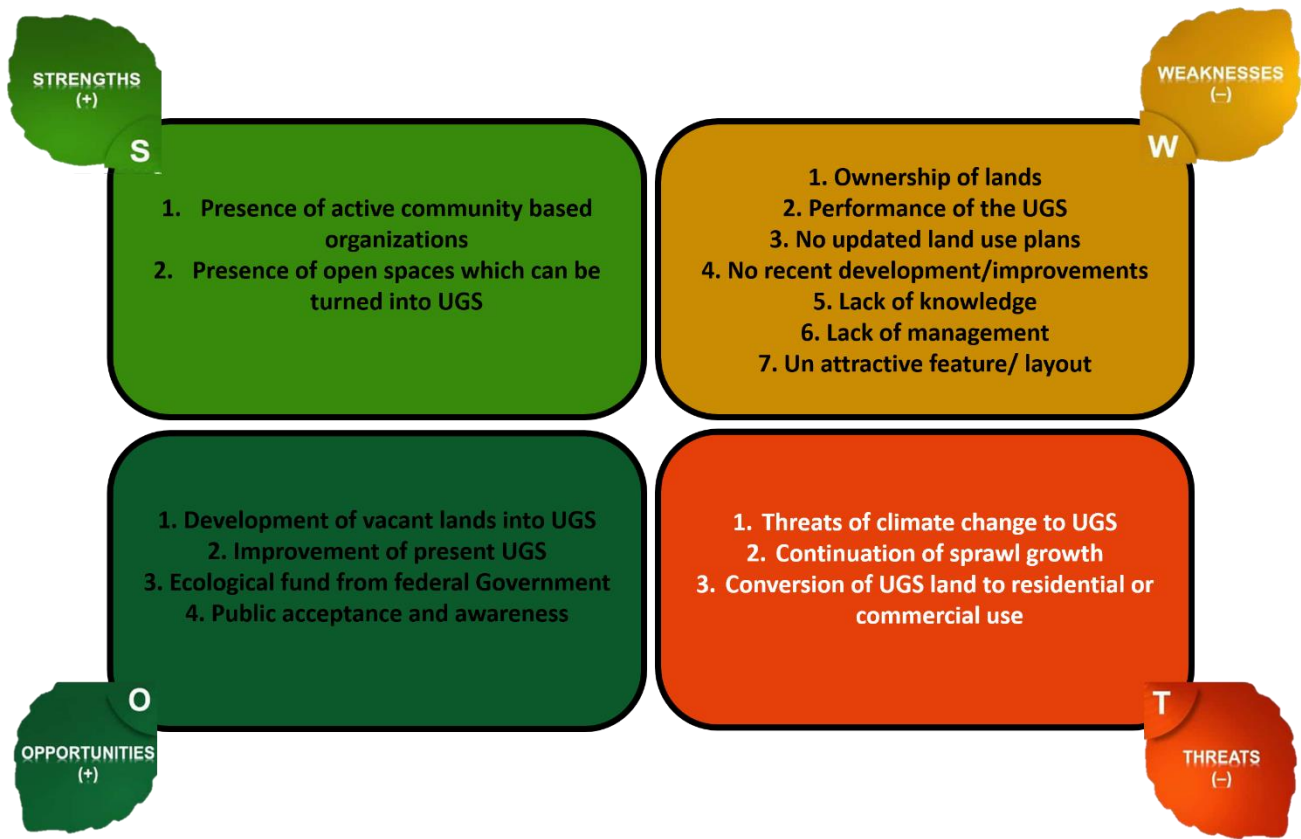
The standard percentage of Recreation in small area which is considered as a ward in this analysis should have 12-14%. Which Is shown in blue color, and the yellow bar is representing the percentage of recreation in study area which is Saidabad. Saidabad has 6.7 in total, which is half of the standard.

S.No	Category	Planning Norms & Standards	
		Population/ unit (Approx)	Plot area (Ha)
1.	Neighborhood Park	10000	1.0
2.	Housing area park	5000	0.5
3.	Tot. Lot at Housing Cluster	350	0.0125

Table Planning Norms, standards for recreational Areas / Parks at neifghborhood

Table 11 Planning Norms, Standards for recreational spaces

3.7. SWOT ANALYSIS



4. RECOMMENDATIONS

Green Roofs

Green Roofs is an innovative yet simple method to improve the existing green percentage of the area. If individuals start roof gardening or Roof plantations it will be Helpful both ways, increase the green percentage and also recreation for the family.



Green roofs are covered with growing media and vegetation that enable rainfall infiltration and evapotranspiration of stored water.

Many green infrastructure elements can be seamlessly integrated into parking lot designs. Permeable pavements can be installed in sections of a lot and rain gardens

and bioswales can be included in medians and along the parking lot perimeter. Benefits include mitigating the urban heat island and a more walkable built environment.

Green streets

Green streets are created by integrating green infrastructure elements into their design to store, infiltrate, and evapotranspire storm water. Permeable pavement, bioswales, planter boxes, and trees are among the elements that can be woven into street design



Land conservation

The water quality and flooding impacts of urban stormwater also can be addressed by protecting open spaces and sensitive natural areas within and adjacent to a city while providing recreational opportunities for city residents.



Assessment of vegetation in green spaces

Depending on the locations, the plant species, names of the main plants grown in major cities of India according to the Gujarat Forest Department report

Table 2. Green spaces and plant species grown in some major cities of India.

City	Green Space (%)	Major Plant Species
New Delhi	11.90	Banyan tree, Peepal tree, Bael, Jamun, Ber, Arjun
Mumbai	6.20	Bhendi tree, Banyan trees, Tamarind tree, Coconut palm, Paral tree, Padauk trees, Mahogany tree, Cajuput tree, Baobab tree, Star Apple, Baobab tree, Peltophorum tree, Gulmohar tree
Chennai	7.50	Sansiveria, Dieffenbachia, Dracaena, Spider plant, Earth star, Money plant, Pothos, Syngonium
Hyderabad	1.66	Bougainvillea, Neerium, Adeneum, Lantana, Euphorbia russelia
Bengaluru	2.96	Australian wattle, Butterfly tree, Red silk-cotton tree, Popcorn bush cedar, Coconut palm, Gulmohur, Indian cork tree, Indian elm, Teak, Silver oak, Orange champak
Kolkata	7.30	Neem, Peepal, Banyan, Radhachura, Krishnachura, Tamarind, Coconut, Betelnut tree
Pune	1.40	Cadamb, Tetu, Awala, Chandan, Tamhan, Muchkund, Cadamb, Kanchan, Putranjiva, Semla Kanchan, Kapila, Murudsheng Ritha, Undi
Panaji	10.00	Coconut, Teak, Sal, Cashew, Mango, Jackfruit tree

The table, above is the types of plants that are suitable for the cities. Hyderabad city can accommodate trees like Neem, Lantana, etc.



BIBLIOGRAPHY

- ❖ Source: Role of Green Spaces for Maintaining Well-Being in Residential Community Development by Prashanti Rao <https://www.intechopen.com/online-first/76802>
- ❖ The relevance of urban green spaces https://www.euro.who.int/_data/assets/pdf_file/0010/342289/Urban-Green-Spaces_EN_WHO_web3.pdf
- ❖ Evaluating the Urban Green Space benefits and functions at macro, meso and micro level: <https://www.ijert.org/research/evaluating-the-urban-green-space-benefits-and-functions-at-macro-meso-and-micro-level-case-of-bhopal-city-IJERTV3IS060563.pdf>
- ❖ Green spaces in cities <file:///C:/Users/90148/Desktop/clg/thesis/FULLTEXT01.pdf>
Analysis and comparison of green spaces in Stockholm and Madrid Mónica Verduras-Stockholm university
- ❖ Green Space.doc <https://wichita.kumc.edu/Documents/wichita/wire/Green-Space.pdf>
- ❖ ARTICLES A Challenge to Cities: How Can We Incorporate Green Spaces? <https://www.arch2o.com/urban-green-spaces-challenge-cities/>
- ❖ Good Practices for Urban Greening <https://publications.iadb.org/publications/english/document/Good-Practices-for-Urban-Greening.pdf>
- ❖ <https://www.iadb.org/> Good Practices for Urban Greening Washington, D.C. May 1997 C ENV-109 - <https://publications.iadb.org/publications/english/document/Good-Practices-for-Urban-Greening.pdf>
- ❖ Urbanization and changing green spaces in Indian cities (Case study - city of Pune): https://www.researchgate.net/publication/264084683_Urbanization_and_changing_green_spaces_in_Indian_cities_Case_study_-_city_of_Pune
- ❖ Urban Green Spaces and Their Need in Cities of Rapidly Urbanizing India: A Review
- ❖ Urban Ecosystem Services Stockholm university <http://su.diva-portal.org/smash/get/diva2:866074/FULLTEXT01.pdf>

- ❖ Source: Urbanization and changing GS in Indian cities (Case study - city of Pune):
https://www.researchgate.net/publication/264084683_Urbanization_and_changing_green_spaces_in_Indian_cities_Case_study_-_city_of_Pune
- ❖ International Journal of Geology, Earth and Environmental Sciences ISSN: 2277-2081 (Online) <http://www.cibtech.org/jgee.html> 2012 Vol. 2 (2) May-August, pp.148-156/Padigala Bhakar
- ❖ Source census of India <http://censusindia.gov.in/pca/pcadata/pca.html>
<https://censusindia.gov.in/pca/pcadata/Houselisting-housing-AP.html>
- ❖ Urban Green Spaces and an Integrative Approach to Sustainable Environment - Shah Md. Atiqul Haq
https://www.researchgate.net/publication/276488760_Urban_Green_Spaces_and_an_Integrative_Approach_to_Sustainable_Environment
- ❖ The Quality of Life and the Environment Ruén Keles
<https://reader.elsevier.com/reader/sd/pii/S1877042812003710?token=09AF535433E271195BF1FA744461215AFE78637B622EA4243B8109CCD0036BC83FBB5D14D12F489D6FD6F4D0CB219D67&originRegion=eu-west-1&originCreation=20210725150212>
- ❖ Good Practices for Urban Greening Washington, D.C. May 1997 C ENV-109
<https://publications.iadb.org/publications/english/document/Good-Practices-for-Urban-Greening.pdf>
- ❖ THE EFFECTS OF SURROUNDING VEGETATION, BUILDING CONSTRUCTION AND HUMAN FACTORS ON THE THERMAL PERFORMANCE OF HOUSING IN A TROPICAL ENVIRONMENT By ALAMAH MISNI
<https://core.ac.uk/download/pdf/41339718.pdf>
- ❖ Urban Green Growth Strategies for Indian Cites
https://www.niua.org/sites/default/files/reports/vol1-12feb2015_web.pdf
- ❖ Sustainable green spaces improve quality of urban living
<https://www.thehindu.com/features/homes-and-gardens/Sustainable-green-spaces-improve-quality-of-urban-living/article12546180.ece>
- ❖ A.P. Urban Greening and Beautification Corporation Ltd. Importance of Urban Greening in the context of climate change

https://aphrdi.ap.gov.in/documents/Trainings@APHRDI/2016/09_Sep/Engineering/Sri%20Prabhakaran.pdf

- ❖ Urban Green Space Interventions and Health
https://www.euro.who.int/_data/assets/pdf_file/0010/337690/FULL-REPORT-for-LLP.pdf
- ❖ Green spaces in cities? Analysis and comparison of green spaces in Stockholm and Madrid
<file:///C:/Users/90148/Desktop/clg/thesis/FULLTEXT01.pdf>
- ❖ Urban Green Spaces and Their Need in Cities of Rapidly Urbanizing India: A Review
Manish Ramaiah 1 and Ram Avtar <file:///C:/Users/90148/Desktop/clg/thesis/urbansci-03-00094.pdf>
- ❖ Assessment of crime and safety issues in parks <http://kth.diva-portal.org/smash/get/diva2:861815/FULLTEXT04.pdf>
- ❖ A Challenge to Cities: How Can We Incorporate Green Spaces?
<https://www.arch2o.com/urban-green-spaces-challenge-cities/>
- ❖ Green Space Management & Residents' Benefits - A Study of Swedish Rental Multi-Family Housing Areas https://pub.epsilon.slu.se/2411/1/Lindgren_T_101119.pdf
- ❖ Planning Practices of Greening Challenges for Public Urban Green Space <http://kth.diva-portal.org/smash/get/diva2:928827/FULLTEXT01.pdf>
- ❖ The role of urban green space for human well-being* Christine Bertram and Katrin Rehdanz
https://www.ifw-kiel.de/fileadmin/Dateiverwaltung/IfW-Publications/Christine_Bertram/the-role-of-urban-green-space-for-human-well-being/KWP_1911.pdf
- ❖ Benefits of Urban Parks A systematic review A Report for IFPRA
<https://worldurbanparks.org/images/Newsletters/IfpraBenefitsOfUrbanParks.pdf>
- ❖ Urban green spaces: a brief for action
https://www.euro.who.int/_data/assets/pdf_file/0010/342289/Urban-Green-Spaces_EN_WHO_web3.pdf
- ❖ Planning strategies and design guidelines for parks open spaces and green areas in Bangalore metropolitan area - Rajan, Sridhar -
<https://shodhganga.inflibnet.ac.in/handle/10603/92399>
- ❖ PLANNING STRATEGIES FOR PARKS, OPEN SPACES AND GREEN AREAS
https://shodhganga.inflibnet.ac.in/bitstream/10603/92399/5/05_contents.pdf